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REVIEW

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Intraosseous administration of antidotes – a systematic review

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ABSTRACT

Context: Intraosseous (IO) access is an established route of administration in resuscitation situations. Patients with serious poisoning presenting to the emergency department may require urgent antidote therapy. However, intravenous (IV) access is not always readily available.
Objective: This study reviews the current evidence for IO administration of antidotes that could be used in poisoning. The primary outcome was mortality as a surrogate of efficacy. Secondary outcomes included hemodynamic variables, electrocardiographic variables, neurological status, pharmacokinetics outcomes, and adverse effects as defined by each article.
Methods: A medical librarian created a systematic search strategy for Medline, subsequently translated to Embase, BIOSIS, PubMed, Web of Science, Cochrane, Database of Abstracts of Reviews of Effects (DARE), and the CENTRAL clinical trial register, all of which we searched from inception to 30 June 2016. Interventions included IO administration of selected antidotes. Articles included volunteer studies, poisoning, or other resuscitation contexts such as cardiac arrest, burns, dehydration, seizure, hemorrhagic shock, or undifferentiated shock. We considered all human studies and animal experiments to the exception of *in vitro* studies. Two reviewers independently selected studies, and a third adjudicated in case of disagreement. Three reviewers extracted all relevant data. Three reviewers evaluated the risk of bias and quality of the articles using specific scales according to each type of study design.
Results: A total of 47 publications (46 articles and one abstract) met our inclusion criteria and described IO administration of 13 different antidotes. These included one case series and 21 case reports describing 26 patients, and 25 animal experiments. Of those, seven human case reports and four animal experiments specifically reported the use of antidotes in poisoning. Human case reports suggested favorable outcomes with IO use of atropine, diazepam, hydroxocobalamin, insulin, lipid emulsion, methylene blue, phentolamine, prothrombin complex concentrate, and sodium bicarbonate. Clinical outcomes varied according to the antidote used. The only reported adverse event was ventricular tachycardia following IO naloxone. Regarding the animal experiments, IO administration of lipid emulsion and of hydroxocobalamin showed improved survival in bupivacaine-poisoned rats and in cyanide-intoxicated swine, respectively. Animal data also suggested an equivalent bio-availability between IO and IV administration for atropine, calcium chloride, dextrose 50%, diazepam, methylene blue, pralidoxime, and sodium bicarbonate. Adverse effect reporting of fat emboli after IO administration of sodium bicarbonate, for example, was conflicting due to the significant heterogeneity in the timing of lung examination across studies.
Conclusion: The evidence supporting the use of IO route for the administration of antidotes in a context of poisoning is scarce. The majority of the evidence consists of case reports and animal experiments. Common antidotes such as acetylcysteine, fomepizole, and digoxin-specific antibody fragments have not been studied or reported with the use of the IO route. Despite the low-quality evidence available, IO access is a potential option for antidotal treatments in toxicological resuscitation when IV access is unavailable.

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Introduction

Antidotes play an important role in the management of intoxicated patients, and sometimes need rapid administration either in a prehospital setting or upon arrival at the hospital, which is why they should be readily available in the emergency department [1]. Most antidotes are administered via the peripheral intravenous route (IV) but obtaining such access might not always be successful in patients with circulatory collapse. In this situation, obtaining central venous access was the common alternative until recently.

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The intraosseous (IO) route is now an accepted safer and faster option than central venous access [2].

Although the use of the IO route dates as far back as World War II [3], it was common in hospitals before the 1980s. Pediatricians have used this route for more than 30 years, and adult resuscitation protocols have recommended this route for over 10 years [4]. The American Heart Association recommends its use in their guidelines while noting a paucity of evidence for effectiveness in the setting of cardiopulmonary resuscitation [5]. Drug monographs rarely discuss pharmacokinetic and pharmacodynamic information for IO administration. However, a review article about IO drug administration during resuscitation can partially offset the lack of information contained in drug monographs [6]. Drugs like adenosine should probably not be given through the IO route. With less than 10 s in blood before being transported into cells for deamination, it would not reach its site of action in time for clinical effect on the cardiac muscle if administered by the tibial IO route, for example.

As the IO route is becoming a more common alternative in emergency situations, knowing the expected effects and safety of certain antidotes given by this route will assist clinicians in critical situations. The goal of this systematic review is to report the available information on antidotes administered by IO route.

Methods

Systematic literature search

A medical librarian (MM) identified candidate studies for this systematic review by creating a systematic search strategy for Medline (via Ovid), which appears in the Appendices. The strategy comprised a combination of Medical Subject Headings (MeSH), title/abstract keywords, truncations and Boolean operators, and included the concepts of antidotes and IO delivery. It was subsequently translated for Embase (via Ovid), BIOSIS (via Ovid), PubMed, Web of Science, Cochrane, Database of Abstracts of Reviews of Effects (DARE), and the CENTRAL clinical trial register. All database searches ran from inception to 30 June 2016 with no limits with respect to language.

Selection criteria

Two reviewers (search 1: AE and LP; search 2: AE and PAD) independently performed the eligibility assessment for each identified study and, where there was still doubt, we obtained the full article to further assess whether the studies met the review criteria. We resolved disagreements between reviewers by discussion with a third reviewer (EV).

The target intervention eligible for inclusion was IO administration of any of the following 32 selected antidotes: acetylcysteine, atropine, calcium chloride, calcium gluconate, calcium EDTA, dantrolene, deferoxamine, dextrose 10% or 50%, diazepam, digoxin immune FAB, diphenhydramine, ethanol, flumazenil, fomepizole, glucagon, hydroxocobalamin, insulin (regular), leucovorin, lipid emulsion, levocarnitine, methylene blue, naloxone, phentolamine, physostigmine, pralidoxime, protamine, prothrombin complex concentrate, pyridoxine, sodium bicarbonate, sodium thiosulfate, thiamine, and vitamin K. We included randomized trials, observational studies, and case reports – in both human and animal, but excluded *in vitro* studies. We also included data presented in abstract form if they contained all the data required. We reviewed editorial comments and narrative reviews only to cross-check their references for further study identification and only included them if containing original data.

Quality assessment and data extraction

Three reviewers (AE, SG, and EV) assessed the methodological quality of animal experiments for risk of bias using the ARRIVE checklist [7], and two reviewers (SG and EV) evaluated human studies with GRADE [8]. Disagreements occurred in the animal experiments and were resolved by using the mean of the three ARRIVE scores calculated by the reviewers. Data extracted from each study included the following: clinical context (such as poisoning, seizure, hemorrhagic shock, and cardiac arrest), intervention, participants, and outcome. The primary outcome was mortality as a surrogate of efficacy. Secondary outcomes included the following: hemodynamic variables (blood pressure and heart rate), electrocardiographic variables (arrhythmias, QRS, and QT intervals), neurological outcomes (mental status), and adverse effects (drug interactions and undesirable effects as defined by each study's authors).

Results

Study selection

The flow diagram of study selection appears in Figure 1. The final analyses included 46 articles and 1 abstract with original data, of which 25 involved animals [9–33] and 22 involved humans [34–55]. Poisoning was the clinical context of antidote use in only four animal experiments and in only seven human case reports [10,12,17,28,36,38,42,45,46,50,55], while 14 experiments involved the use of healthy animals [13–16,19,21–26,29,31,32]. Table 1 summarizes the evidence with the number of reports in each category.

Antidotes

Of the 32 antidotes included in the systematic search strategy, only 13 appeared in the publications retrieved during the literature search. We found no data for intraosseous use of the following 19 antidotes: acetylcysteine, calcium gluconate, calcium EDTA, dantrolene, deferoxamine, digoxin immune FAB, diphenhydramine, ethanol, flumazenil, fomepizole, glucagon, leucovorin, levocarnitine, physostigmine, protamine, pyridoxine, sodium thiosulfate, thiamine, and vitamin K. Table 2 summarizes the main results, while details on each study appear in the Appendices.

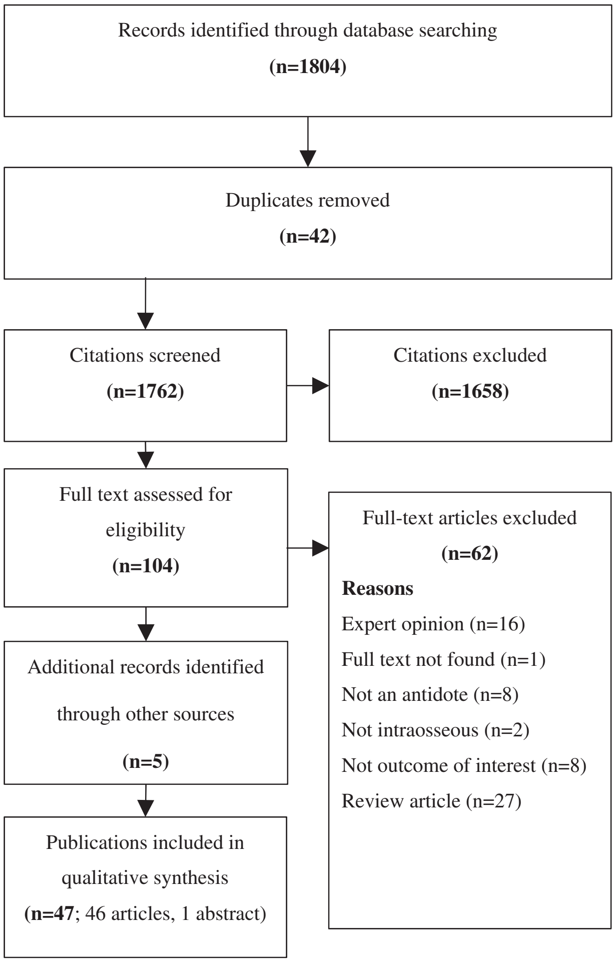

Figure 1. Selection of articles flow diagram. Final search date: June 30, 2016.

Table 1. Summary of evidence regarding IO administration of antidotes.

	Animal studies		Human reports		
Antidote	Total	Poisoning model	Case reports	Case series	Poisoning cases
Atropine	4		7	1	1
Calcium chloride	2				
Dextrose	3		2		
Diazepam	3	2	1		
Hydroxocobalamin	5	1	1		1
Insulin			3		
Lipid emulsion	1	1	2		2
Methylene blue	1		1		1
Naloxone			1		1
Phentolamine			1		
Pralidoxime	2				
PCC			1		
Sodium bicarbonate	11		12	1	1

Atropine

One case series reported the use of IO atropine in the emergency department [43]. The main indication for IO use in this study was cardiorespiratory arrest. That study showed no evidence of immediate complication and reported a positive hemodynamic response to atropine within one minute in two out of two patients. Additionally, six human case reports described the use of IO atropine in adult or in children in different clinical contexts [37,44,46,48,52,53], one of which reported the use of IO atropine in a 2-year-old infant intoxicated with the liquid flea-killer malathion [46].

Three animal experiments measured the systemic bioavailability of atropine when administered by different routes such as IO, intravenous (IV), intramuscular (IM), and endotracheal [21,26,33]. One of them was conducted on minipigs, with the authors concluding that the area under the curve (AUC) was similar among the three routes (IO, central IV, IM), while mean time to achieve maximum concentration (T_{max}) was the same between the IO and central IV routes (2 min) but longer with the IM route (3.5 min). However, the experiment lacked sufficient power to detect a statistical difference among the three treatment routes, as there were only 4–8 minipigs per group [21]. Similarly, Yost et al. looked at the kinetics of atropine when given intraosseously, by the peripheral IV route and by the IM route in euvolemic and hypovolemic swine. The data concluded that T_{max} was immediate for IO and peripheral IV routes (0 min) in euvolemic and hypovolemic swine whereas the IM route required a longer time (6 min in normovolemic swine and 19.5 min in hypovolemic swine). The AUC was also similar between the three routes for both volemic statuses [33]. In the animal experiment by Prete et al., with pigtail macaques, the authors observed a faster average Tmax with peripheral IV (1.37 min) than with the IO route (3.87 min) and than with the endotracheal route (9.62 min). They also observed a slightly greater estimated AUC with IO compared with IV [26].

Finally, one animal experiment studied the risks of fat and bone marrow emboli to the lungs following IO administration of emergency drugs such as atropine to healthy dogs [23]. The researchers counted emboli per square millimeter of area, on slides cut from representative sections of lungs extracted from sacrificed animals at the end of the experiment; fat and bone marrow emboli to the lungs were found in all emergency drug groups. There was no significant difference in the mean number of emboli/mm^2 of lung among the emergency drugs and compared with controls (IO 0.9% NaCl). Moreover, no decrease in PaO_2 occurred during the 4-h study period. Atropine was given with epinephrine and sodium bicarbonate and not as a single drug. The doses used in these animal experiments are comparable with those used in humans.

Calcium chloride

Orlowski et al. [23,24] studied intraosseous calcium chloride in two animal experiments. In the first, they evaluated the safety of IO administration of different emergency drugs and solutions in dogs sacrificed to quantify the bone marrow and fat emboli/mm^2 of lung that resulted from the injection [23]. Three dogs received 10 mg/kg bolus of calcium chloride, a similar dose to that used in pediatrics. No ventilation-perfusion abnormalities were shown after the IO administration, since the arterial blood gas changes were insignificant. Autopsy found emboli in all animals, with no statistically significant difference between the treatment and control groups. Autopsy found no abnormalities at the needle insertion site of femur bone marrow. Second, they compared the

Table 2. Summary of data regarding IO administration of antidotes.

Antidote	Type of study	Doses	Endpoints	Outcome
Atropine	2 animal RCE [21,33] 1 observational cross over study [26]	0.01 mg/kg to 0.12 mg/kg and 2 mg fixed dose (not weight based)	Safety [23]	No significant difference in mean number of fat and bone marrow emboli to the lungs (emboli/mm^2) between the emergency drugs and when compared with the control group [23]
	1 controlled animal experiment [23]		Kinetics [21,26,33]	Similar bioavailability between IV (peripheral and central) and IO groups [21,26,33] Longer T_{max} when compared with peripheral IV [26]
	Case series/reports [37,43,44,46,48,52,53]	0.005 mg/kg, 0.1 mg or 1 mg, and in most cases not mentioned	Safety [43,44,46,52]	No immediate complication reported [43,52]
			Hemodynamics [43,53] Mortality [37,43,44,46,48,52,53]	No delayed complication reported [44,46] Improved bradycardia [43,53] Intact survival [44,46,53] Death [37,43,48,52]
Calcium chloride	1 animal RCE [24] 1 controlled animal experiment [23]	10 mg/kg	Safety [23]	No significant difference in mean number of fat and bone marrow emboli to the lungs (emboli/mm^2) between the emergency drugs and when compared with the control group [23]
			Kinetics [24]	Similar kinetics parameters (T_{max}, C_{max} between IV (peripheral and central) and IO groups [24]
Dextrose	1 animal RCE [24] 2 controlled animal experiments [22,23]	0.25 g/kg to 0.5 g/kg	Safety [23]	No significant difference in mean number of fat and bone marrow emboli to the lungs (emboli/mm2) between the emergency drugs and when compared with the control group [23] Second greatest bone marrow damage at insertion site [23]
			Kinetics [22,24]	Similar kinetics parameters (T_{max}, C_{max}) between IO and central IV access [22,24] Statistically lower C_{max} for peripheral IV [24]
	Human case reports [50,51]	0.5 g/kg diluted or unrecorded volume	Mortality [50,51]	Survival [50,51]
Diazepam	1 animal RCE [15] 2 controlled animal experiments [10,28]	0.1 mg/kg to 3 mg/kg and 4 mg fixed dose	Neurological outcomes [10,28]	Cessation of seizure activity on EEG [10,28]
			Kinetics [10,15,28]	Similar time to control seizure activity Similar blood concentration over time between peripheral IV and IO groups [10,15,28]
	Human case report [48]	0.125 mg/kg	Neurological outcomes [48]	Cessation of seizures [48]
Hydroxocobalamin	4 animal RCE [11,12,14,21] 1 prospective pilot study [20]	35 mg/kg to 150 mg/kg	Mortality [12,14,20]	Mortality rate similar between IV and IO groups [12] Intact survival [14,20]
			Hemodynamics [11,12,20]	Same between IV and IO groups [12] Same between IO blood infusion and IO and IV hydroxocobalamin infusion [11] Blood pressure increased with increasing doses of IO hydroxocobalamin [20]
			Kinetics [21] Safety [12,14]	Similar bioavailability between the central IV and the IO group [21] No difference in complications between IV and IO groups [12,14]
	Human case report [36]	2.5 g	Mortality Hemodynamics	Survival without sequelae HR and SBP stabilization
Insulin (regular)	Human case report [34,39]	0.1 unit/kg/h × 10 hours and decreased to 0.08 unit/kg/h × 4 hours (total duration of IO infusion 14 h) and	Mortality [34,39] Safety [34,39]	Survival [34,39] No immediate complications [34,39]

(continued)

Table 2. Continued

Antidote	Type of study	Doses	Endpoints	Outcome
		unrecorded dose of insulin infusion; total duration of IO infusion 12 h		
Lipid emulsion	1 animal RCE [17]	10 ml/kg of lipid emulsion 30%	Mortality Hemodynamics	Same survival between IV and IO groups Similar rate of hemodynamics recovery
	2 human case reports [38,55]	12 ml bolus followed by an infusion of lipid emulsion 20% (infusion rate and duration not reported) and unrecorded lipid emulsion volume	Mortality [38,55] Neurological outcomes [38]	Intact survival with no further seizures [38] Death [55]
Methylene blue	1 animal experiment [19]	10 mg (10 ml) over 20 s	Kinetics	Similar entry time into the aorta between peripheral IV and IO groups
	Human case report [42]	1 mg/kg (0.3 ml) over 3–5 min	Mortality	Survival without sequelae
Naloxone	Case series [45]	0.4 mg push	Mortality Hemodynamics	Survival despite ICU admission for hemodynamics instability Ventricular tachycardia, shock
Phentolamine	Human case report [41]	5 mg push	Mortality Hemodynamics	Survival and salvage of the limb from vasopressor extravasation Return of palpable dorsalis pedis pulse
Pralidoxime	2 animal RCE [21,30]	25 mg/kg to 660 mg fixed dose	Kinetics	Similar bioavailability between IV and IO groups [21] Significant higher Cmax and faster Tmax with IO route compared with IM route [30]
Prothrombin complex concentrate	Human case report [49]	IO Profilnine SD (factor IX complex [human] [factors II, IX, X]) 1490 U (33 U/kg) reconstituted with 10 mL of manufacturer provided diluent and administered at its maximum rate of 10 mL/min	Mortality Safety Hemodynamics	Survival No complications besides local pain during injection Systolic blood pressure improved
Sodium bicarbonate	6 animal RCE[a] [16,18,24,25,27,32] 2 controlled animal experiments [23,28] 3 animal experiments [9,13,31]	1 mEq/kg [16,23,24,25,27,28,31,32] 2 mEq/kg [9,13,18] Infusion: 10% $NaHCO_3$ solution in D5% at 80 ml/kg/d [25] to 100 ml/kg/d for 1 h [32]	Safety [13,16,18,23,25,28,32]	No significant difference in mean number of fat and bone marrow emboli to the lungs (emboli/mm2) between the emergency drugs and when compared with the control group [23] Fat globules in the peribronchial blood vessels and perialveolar region with no difference in distribution and percentage with CPR alone [18] Greatest bone marrow damage at insertion site [23] 4 h post infusion No histological, microscopic, radiologic and bone scan differences between saline and $NaHCO_3$ infusions at 3 d, 5 d [16], and at 30 d [28]. No bone growth disturbance observed at 3 weeks [13], 3 months [25], nor at 6 months [32] Complete epiphyseal closure at 6 months [32] Similar marrow cellularity between experiment and control limbs at 3 months [25] Similar kinetics parameters between central IV and IO groups [27] Similar kinetics parameters between peripheral IV and multiples IO sites except for malleolar site, Tmax significantly longer [31] Longer T_{max} and longer duration of effect compared with peripheral and central IV [24]

(*continued*)

Table 2. Continued

Antidote	Type of study	Doses	Endpoints	Outcome
			Kinetics [9,24,27,31]	IO bicarbonate injections have different effects on the pH value of IO site and central venous blood; pH and PCO2 of IO blood were higher and clinically different compared with mixed venous site and contralateral IO site at 10, 15, and 20 min of cardiopulmonary resuscitation [9]
	Case series [43]	As variable as:	Mortality	Death for all adult patients [43] and 3 infants in cardiopulmonary arrest [47,48] and for 2 newborns in shock [40,52]
	12 human case reports [35,40,42,47,48,50,51,52,54]	Not mentioned [40,43,52], 10 mL [47], 2 mEq/kg [42], 8.4% 2 mL/kg [51], 6 mEq [48], 2,6 mEq/kg [50], 1.4% 40 mL/kg [35], 1.4% 15 mL/kg [54], 1.4% 25 mL/kg (40 mL, 1 mEq/kg) [54]		Intact survival in 6 infants (dehydration [35,48,54], dehydration and poisoning [42], poisoning [50], anaphylaxis/cardiac arrest [51]) Wide QRS and ventricular fibrillation to sinus rhythm and QRS normalization [50] General improvement of hemodynamics [54] Recovery of consciousness [50]
			Electrocardiographics and hemodynamics [50,54] Neurological outcomes [50] Safety [35,42,43,52,54]	No evidence of complications from IO access [35,42,43,52,54]

[a]RCE: randomized controlled experiment.

effectiveness and distribution of different emergency drugs and solutions when administered by central IV, peripheral IV, and by the IO route [24]. Three dogs received 10 mg/kg bolus of calcium chloride by each administration route with analysis of ionized calcium concentrations to determine the drug distribution. There was no statistical significant difference in changes of calcium concentration over time in dogs among the three routes of calcium administration; the effectiveness of calcium IO administration was similar to its IV administration.

Dextrose 50%

In humans, there are only two case reports with the use of IO dextrose 50%. The two cases both described the administration of IO dextrose 50%, sodium bicarbonate, and other resuscitation drugs in different contexts, such as encainide intoxication [51] and cardiac arrest [52] in an infant. However, the outcomes described are not related to the dextrose 50% administration and are thus of poor value.

Two animal experiments, one on 14 swine and the other on 21 dogs, looked at the change in serum glucose after a bolus of hypertonic glucose (dextrose 50%) via different administration routes such as central and peripheral IV line and via IO route [22,24]. The results of both experiments showed similar blood glucose concentration over time, and similar time to achieve maximal glucose concentration between central IV and IO. The experiment on dogs also compared the peripheral IV route to the IO and central IV routes. Time to peak (T_{max}) was the same but the peak concentration (C_{max}) was statistically lower in the peripheral IV route than in the two other routes [24]. These experiments included small animal samples – only 10 and three animals received dextrose 50% by the IO route. Additionally, the experiment on swine was not randomized, and baseline serum glucose concentrations differed statistically between the two groups [22]. Finally, another animal experiment studied the risks of fat and bone marrow emboli to the lungs and bone marrow necrosis following IO administration of emergency drugs to healthy dogs, in which only three subjects received dextrose 50% [23]. Fat and bone marrow emboli to the lungs were present in all emergency drug groups with no significant difference in mean number of emboli/mm^2 of lung among the emergency drugs and compared with controls (IO 0.9% NaCl). The PaO_2 decrease of 27.1 mmHg in the arterial blood gases that occurred during the 4-h study period, as PAO_2-PaO_2 and intrapulmonary shunt, did not constitute a significant ventilation-perfusion alteration according to the authors. The authors did not report a clinically acute pulmonary distress syndrome or adult respiratory distress syndrome secondary to emboli. The examination of the femoral insertion site revealed that the second greatest bone marrow damage (necrosis and hemorrhage) after sodium bicarbonate infused alone or combined with epinephrine occurred with infusion of dextrose 50%. The doses used in these animal experiments are comparable (0.5 g/kg) or inferior (0.25 g/kg) to the one used in humans to treat hypoglycemia.

Diazepam

One human case report described the administration of IO diazepam in a 6-month-old infant with seizures. According to this case, the seizures stopped 1 min after the proximal tibia IO administration [48].

Three animal experiments studied the use of IO diazepam, two in an pentylenetetrazol-induced seizure model (swine and piglets) [10,28] and one in healthy dogs [15]. Concerning kinetics outcomes, both Brickman et al. and Spivey et al. showed a slightly higher diazepam blood concentration during the first 5 min in the IV group compared with the IO group although this was not statistically significant [15,28]. Beall et al. in an abstract similarly observed no statistical differences in the maximal blood diazepam concentration between IO and IV routes [10]. Regarding the time to control seizure activity, the Spivey et al. swine seizure model did not show significant difference between IV or IO route. However, the mean time to complete suppression of seizure activity was 2 min in the IV group and 8 min in the IO group [28]. Beall et al. also reported a cessation of EEG activity in the seizing piglets with the IO and IV routes without mentioning specific times in the abstract [10]. The doses used by Spivey et al. are comparable to doses used to treat a status epilepticus in pediatrics patients [28]. The 4 mg fixed dose of the Beall experiment would be similar to the dose used in a 13 kg pediatric patient (0.3 mg/kg) to treat status epilepticus.

Hydroxocobalamin

A single human case report described a 9-month-old infant who received IO 2.5 g hydroxocobalamin for presumed smoke inhalation cyanide poisoning in a prehospital setting. Her hemodynamic values stabilized after antidote administration, and she was discharged on day 9 of hospitalization without any neurological or other sequelae [36].

Five animal experiments included two healthy animal models (goats and minipigs) [14,21] and three with swine in the context of poisoning or hemorrhagic shock with doses of 35 to 150 mg/kg IO hydroxocobalamin [11,12,20]. The kinetic outcomes, mortality, and hemodynamic parameters were similar between IO and IV administration. More precisely, in severe hemorrhagic models, the positive outcomes on SBP, MAP, SVR, and HR were similar among IO hydroxocobalamin, IV hydroxocobalamin, and IO whole blood groups, and superior to the control/no treatment group. Meanwhile, serum lactate improved in all treated groups but deteriorated in the control group [11]. The animal experiment models are similar to human poisoning and hemorrhagic scenarios. Doses were comparable with those recommended in human clinical poisoning.

Insulin (regular)

We found three human case reports describing the use of insulin by the IO route in pediatric diabetic ketoacidosis (DKA). A 5-year-old child with severe DKA received an insulin infusion (0.1 unit/kg/h) [34]. However, doses of insulin used in the treatment of DKA are lower than doses recommended in cardiovascular poisoning from calcium channel blocker or beta-adrenergic antagonists. The other two pediatric cases included in Gallo et al. did not specify the doses used [39]. No complications associated with the IO insulin infusion were reported. We found no study in the literature describing the use of insulin by the IO route in an animal model.

Lipid emulsion

One human case report briefly described the successful use of IO lipid emulsion in an infant with tonic–clonic seizures from lidocaine toxicity [38]. Another human case report described the use of IO lipid emulsion in an adult with verapamil overdose; at first, it was very painful for the patient, then it could not be completed through the IO route due to difficulties with flow resistance via the pump delivery system [55]. Although the patient died 2 d later and issues with IO administration occurred early, the authors stated that lipid emulsion can be attempted by the IO route.

One animal experiment with bupivacaine-intoxicated rats reported similar mortality and hemodynamic outcomes between IO and IV administration [17]. The lipid emulsion administered was 10 times the recommended dose used in human clinical poisoning.

Methylene blue

Two articles reported methylene blue IO administration, an animal experiment and a pediatric case report. Herman et al. [42] reported a human case of methylene blue IO administration. A 6-week-old female infant with methemoglobinemia received a slow IO infusion of 1 mg/kg methylene blue in an emergency department. The severe cyanosis resolved after 8 min while oxygen saturation increased from 86% (when receiving oxygen 10 L/min) to 98–100%. Methemoglobin decreased from 29.8% to 8.2% over the 3 h following the IO administration. Her physicians then administered a second dose of methylene blue by the IV route. No apparent adverse effect resulted from IO administration of methylene blue, saline, and sodium bicarbonate to correct both methemoglobinemia and metabolic acidosis; the infant recovered and was discharged after 3 d of hospitalization.

Hosseinpour et al. conducted an experiment involving 20 rabbits (1400–2600 g) that received either peripheral IV or IO administration of 10 mg/10 mL of methylene blue. The drug delivery into the systemic circulation was evaluated by the time of methylene blue appearance in the aorta as seen by the naked eye [19]. Rabbits received 4–7 mg/kg of methylene blue, which is higher than the doses used clinically (1–2 mg/kg). All rabbits survived until the end of the experiment. There was no statistical significant difference found in the entry time of methylene blue in the aorta whether administered by IV or IO routes.

Naloxone

One human case report described a naloxone IO administration in a multi-drug user adult suspected of overdose. Presenting with an altered mental state and a history of prior methadone and drug overdoses, the paramedics administrated a protocol of two IO doses of naloxone 0.4 mg. Once in the emergency department, he was in ventricular tachycardia at a rate of 160–180 BPM. Ventricular tachycardia is a rare complication reported with naloxone administration. He first received magnesium sulfate and amiodarone then cardioversion. The patient had multiple risk factors for dysrhythmia (congenital heart disease, chronically high opiate doses, and multi-drug user). He also had tachypnea, respiratory alkalosis, a low Glasgow coma score, and a progressive agitation leading to sedation and intubation. The patient's condition stabilized but required a transfer to the intensive care unit [45]. We found no study in the literature describing the use of naloxone by the IO route in an animal model.

Phentolamine

We found only one human case report in the literature for phentolamine [41]. It described the use of IO phentolamine in an adult with extravasation of dopamine and/or norepinephrine through his IO route. The treatment consisted of phentolamine 5 mg into the IO needle, nitroglycerin 2% ointment, and injection of verapamil 2.5 mg and nitroglycerin 0.2 mg in the femoral artery. This treatment resulted in salvage of the patient's limb. We found no study in the literature describing the use of phentolamine by the IO route in an animal model.

Pralidoxime

We found no human case report in the literature. One animal experiment conducted on minipigs examined the systemic bioavailability of pralidoxime when administered by different routes such as IO, IV, and IM [21]. The authors concluded that the area under the curve was similar among the three routes. Also, the mean time to achieve maximum concentration was the same between IO and central IV route (1.75 min). The dose used in this animal experiment is comparable with the dose used in humans. This experiment was not powered to detect a statistical difference among the three treatment routes, as there were only 4–8 minipigs per group. Another animal experiment on swine found that IO delivery of pralidoxime at standard doses provided faster delivery to the circulation compared to IM delivery [30]. In addition, an IO infusion, followed by IM injection up to 1 h later, may maintain the therapeutic range of pralidoxime for a prolonged length of time.

Prothrombin complex concentrate (PCC)

In the first human case report of a PCC product administered IO for gastrointestinal hemorrhage, the same dose that would be used by the IV route was given to the patient by the IO route without adverse effects [49]. The patient's clinical status stabilized and the melena subsided, but laboratory monitoring could not confirm that the PCC product contributed to controlling the bleeding. We found no study in the literature describing the use of PCC product by the IO route in an animal model.

Sodium bicarbonate

A case series reported IO use of sodium bicarbonate in seven adults with cardiac arrest. They received emergency drugs and solutions via an IO access, including sodium bicarbonate (doses unknown), with an increase in pH in four cases and without evidence of complications from IO access. No evaluation was possible for the pH variation in the three other cases as only one blood gas sample was taken during resuscitation [43].

In a pediatric population, 12 cases reported IO use of sodium bicarbonate, but only one for antidotal therapy. Following accidental ingestion of 25 mg of encainide, a 6-month-old boy presented a wide-complex sinus tachycardia evolving to ventricular tachycardia without a pulse. His condition deteriorated despite cardioversion and defibrillation. His doctors administered IO phenytoin via the tibial route, dextrose and sodium bicarbonate 2.6 mEq/kg (used as antidote to reverse the sodium-blocking effect of encainide), and his condition improved: a palpable pulse was noted with BP of 108/63 mmHg, eyes opening, agitation, and return to sinus rhythm with a normal QRS interval [50].

Five pediatric case reports describe the use of IO sodium bicarbonate in cardiorespiratory arrest. A 3-month-old male died the day after his arrival at the emergency department where he received emergency drugs via endotracheal access and 10 mL sodium bicarbonate and fluids via IO access [47]. A 5-month-old female and her 17-month-old brother did not recover from smoke inhalation, despite ventilation and IO epinephrine, atropine and bicarbonate [48]. An 18-month-old male with presumed penicillin anaphylaxis and secondary cardiorespiratory arrest received tibial IO administration of epinephrine resulting in a return of spontaneous circulation. The infant survived with no complications noted at the medical follow-up 3 months later [51]. A preterm female born at 28 weeks of gestation needed venous access for medication administration after pulmonary function deterioration on her 38th day of life. Her doctors stabilized her with tibial IO administration of albumin, epinephrine, atropine, sodium bicarbonate (no information on doses), dextrose, antibiotics, dobutamine, and fentanyl. Her ventilatory condition, electrolytes and blood gases' results improved, but the infant still died from an acute respiratory failure 24 h later. However, no abnormality at the administration site or at the bone marrow space was noted on autopsy [52].

Four case reports of infants (a 7-month-old, a 5-month-old, and two 9-month-olds) suffering from severe dehydration secondary to gastro-enteritis or isolated diarrhea and fever received IO sodium bicarbonate (6 mEq, 1 mEq/kg, 1.4% 40 mL/kg, and 1.4% 15 mL/kg). The first case also received IO fluids, IV antibiotics, and red cell transfusion prior to recovery [48]. The second case also received colloids and albumin via tibial IO access. His mental status improved. His pH and sodium concentrations were 7.31 and 162 mEq/L, respectively, after sodium bicarbonate administration [35]. The third and fourth patients presented improvement of hemodynamic parameter and neurologic status (responsiveness in the third patient, seizure cessation in the fourth patient after phenobarbital administration). Improvement in the electrolytes values was also mentioned in the third case. Both the third and fourth patients received other drugs and solutions via the IO route. The second infant had no abnormalities noted at IO site at hospital discharge. The third infant had no tibial growth problem at 1-year follow-up. The fourth infant had no tibial anomaly at day 15 of hospitalization [54].

One case report of a 15-d-old newborn with septic shock who received an unknown dose of IO sodium bicarbonate and epinephrine reported a transient increased heart rate but the infant ultimately died [40].

Herman et al. described a pediatric case report of acquired methemoglobinemia on dehydration. The 6-week-old infant received IO sodium bicarbonate 2 mEq/kg to correct metabolic acidosis (pH 7.21 before treatment). This case appears in the methylene blue section [42].

Finally, seven animal experiments evaluated the safety of IO sodium bicarbonate infusion [13,16,18,23,25,28,32]. Two assessed the risk of fat embolisms [18,23] and five the risk of bone growth defect [13,16,25,29,32]. Four animal studies evaluated kinetic outcomes [9,24,27,31]. The results appear in the following sections.

Fat embolisms. Two animal experiments examined the risk of fat and bone marrow emboli to the lung after IO administration of different emergency antidotes such as sodium bicarbonate. Orlowski et al. [23] quantified the bone marrow and fat emboli/mm^2 of lung in healthy dogs. They found fat and bone marrow emboli to the lungs in all animals, including controls who received a saline solution. The group of dogs who only received IO sodium bicarbonate had the fewest number of emboli whereas the group who received the combination of IO epinephrine and sodium bicarbonate had the greatest number of lung emboli. However, the statistical analysis showed no significant difference of mean number of emboli/mm^2 of lung or in arterial PaO_2 between the groups. Samples were collected four hours post infusions and thus reflected early effects of IO infusions. Finally, the examination of the femoral insertion sites revealed that the greatest bone marrow damage (necrosis and hemorrhage) was after sodium bicarbonate administration [23]. The dose of 1 mEq/kg of undiluted sodium bicarbonate used, without further concentrations described, is comparable with the dose used in humans.

Fiallos et al. conducted the other experiment on piglets and assessed the incidence and magnitude of fat emboli after cardiopulmonary resuscitation (CPR) and IO infusions in a cardiac arrest model. Knowing that CPR alone can produce fat and bone marrow emboli, this study assessed the risk of CPR and an IO infusion combination. Thirty-three piglets underwent CPR and were randomized to receive or not to receive IO sodium bicarbonate with CPR. All animals had fat globules in the peribronchial blood vessels and perialveolar region with no differences in the distribution and appearance between the groups [18]. These results suggested that IO bicarbonate infusion did not increase the incidence and magnitude of fat emboli to the lung compared with CPR alone. The dose of sodium bicarbonate used in this experiment (2 mEq/kg ×3 doses) can be comparable with doses used in toxicological resuscitation in humans, especially with cyclic antidepressant as stated in the most recent ACLS guidelines for special populations [56].

Bone growth and bone damage. Bielski et al. and Dedrick et al. both used an immature rabbit animal model to simulate the effect of IO infusions on human infant bone growth and growth plates [13,16]. Both experiments used saline or sodium bicarbonate. In one experiment, the animals were divided in two groups, with bone specimen examination 24 h and 3 weeks after IO infusion. None of the tibias showed evidence of growth disturbance, and the trauma (well-defined puncture holes and hematomas) to the metaphyseal bone at 24 h was completely resolved at 3 weeks [13]. The other experiment looked at bone injury 3 and 5 d after tibial IO infusions. There were no radiographical and histological differences between saline and sodium bicarbonate groups [16]. The doses of sodium bicarbonate used in these animal experiments are comparable with the doses used in children and were given either undiluted or diluted up to 1 mL (initial and final concentrations not mentioned). Three other animal experiments assessed long-term bone damage in swine model [25,29,32]. Pollack et al. and Woodall et al. both randomized 18 swine to receive IO infusions of fluids or resuscitation drugs, including sodium bicarbonate undiluted bolus or 1-h infusion of a 10% solution in D5% at 80 to 100 mL/kg/d. They examined tibia radiographs at 3 and 6 months post IO infusions. They also analyzed marrow cellularity and white and red blood cell counts by bone marrow aspiration at 3 months. The other tibia of each animal provided the controls. At 3 months, there was no evidence of osteomyelitis, bone defects or any perceptible differences in marrow cellularity and cell precursors' counts between the treated and control tibias [25]. At 6 months, the results showed no difference in linear growth and in tibia length and showed complete epiphyseal closure [32]. Spivey et al. compared the effect of sodium bicarbonate infusion with IO saline infusion on bone integrity at 30 d. All five swine received saline infusion in one tibia and sodium bicarbonate in the other tibia. They obtained radiographs, bone scans, and microscopy of each tibia. Only one animal who received sodium bicarbonate had increased radiodensity of the bone cortex at the IO site on the radiograph at 30 d. Otherwise, they found no evidence of osteomyelitis on any scan or any microscopic abnormalities. There was no difference between the saline and the sodium bicarbonate tibias [29]. The bolus doses used in these three animal experiments are comparable to the doses used in humans but the studies' infusion rates were higher than in clinical practice.

Kinetics. Four animal experiments reported on kinetics outcome of IO sodium bicarbonate infusion. The first three experiments used a 1 mEq/kg dose of sodium bicarbonate. Spivey et al. randomized 23 domestic swine in induced cardiac arrest to compare blood pH following the administration of sodium bicarbonate by IO, central IV, and peripheral IV routes. They performed CPR immediately at the onset of cardiac arrest. Femoral arterial blood pH did not significantly differ between IO and central IV routes during the whole experiment. As for the peripheral IV route, pH was significantly lower than after IO and central IV administration 4 min after the beginning of the infusion and remained lower for the rest of the experiment (30 min). The IO route was the slowest route to reach peak pH (4 min) compared with central IV and peripheral IV (2 min) [27]. Orlowski et al. performed a similar experiment on healthy dogs. They compared pharmacokinetics of six different emergency drugs, such as sodium bicarbonate, given by the same three different administration routes. They studied the change in end-tidal carbon dioxide percentage following sodium bicarbonate infusion. Similarly, as reported by Spivey et al., the time to reach the peak effect was slowest for the IO route. Injection times were significantly different between the three routes and longest in the IO group. The height and peak level of end-tidal carbon dioxide with the IO route was between the central and peripheral route values. The longest duration of effect on end-tidal carbon dioxide was in the IO group [24]. Third, Warren et al. did a prospective descriptive study on intubated and mechanically ventilated piglets to assess sodium bicarbonate response following administration via different IO sites (three of the four following sites in each animal: distal femur, humerus and tibia, or malleolus), as well as from peripheral IV administration. Once again, they also studied the change in end-tidal carbon dioxide following each sodium bicarbonate infusion, with sufficient time between each injection to allow end-tidal CO_2 to return to baseline values. They noted no difference between sites for the time to initial end-tidal CO_2 elevation (average: 12.8 s) or for the maximal end-tidal CO_2 elevation (average: 1.09 kPa). With a post-hoc analysis determination of divergence, the time to maximal end-tidal CO_2 elevation appeared slower after IO than IV injection across all sites, but statistically different from malleolar IO site only. Authors mentioned that those differences "may be partly explained by travel distance, dispersion within bone and lower uptake for intramedullary vessels" [31]. However, the differences would not be clinically significant and the authors considered them negligible [31]. Finally, Abdelmoneim et al. [9] conducted a prospective randomized study with 32 piglets to assess if IO blood samples during prolonged cardiopulmonary resuscitation were as reliable as central venous blood samples to determine central acid–base balance. Twelve animals received IO sodium bicarbonate 2 mEq/kg/dose; six of them received three doses at 5 min of interval, and the six others under IO normal saline perfusion received two doses of IO epinephrine at 5 and 10 min and one dose of IO sodium bicarbonate 2 mEq/kg at 15 min. In the animals who received only IO sodium bicarbonate, pH and PCO_2 of IO blood were higher compared with mixed venous site and contralateral IO site at 10, 15, and 20 min of cardiopulmonary resuscitation. With three doses of bicarbonate, the mean pH values increased to 7.43 in IO site, 7.28 in contralateral IO site and 7.38 in central venous site at 30 min, while the mean pH in the other groups remained between 6.90 and 7.23 (except one mean value of 7.37) in all confounded sites. In the animals that received epinephrine and bicarbonate, they noted no statistical difference in pH values between sites, except after 30 min where central venous site pH was at least 0.1 unit lower than IO pH values. The pH increased to 7.52 in IO blood 5 min after bicarbonate infusion compared with a pH value of 7.07 in mixed venous blood. Since IO bicarbonate injections increased the pH value between the IO and central venous blood differently, the authors concluded that "during prolonged cardiopulmonary resuscitation, drug infusions may render the IO site inappropriate to judge central acidosis" [9].

Discussion

In this review, only 13 antidotes among the 32 included in our search had information on IO administration. A minority of the animal experiments evaluated IO administration in a poisoning model. A similar minority of human cases involved treatment of poisoning. Nevertheless, this information suggests that IO administration of these antidotes is feasible, probably safe, and possibly effective in poisoning.

Many of the studies retrieved in this review appeared more than 20 years ago and often by the same authors. This could lead to lack of generalizability if the medical management of poisoning or the criteria for the diagnosis had changed over the years, which is not the case for most of the antidotes.

In terms of the possible complications from IO insertion, fat embolism is one of the main concerns. The evaluation and the diagnosis criteria of this syndrome have changed little over the years. Therefore, despite the large time span between the publications of the included studies, the data obtained are still relevant in today's clinical practice. As sodium bicarbonate is sometimes used in toxicological cardiopulmonary resuscitation, more case reports and trials relating its use and its safety via the IO route are available. CPR can produce fat and bone marrow emboli by itself. Fiallos et al. [18] found no difference in incidence of pulmonary fat emboli between sodium bicarbonate and CPR or CPR alone. These findings tend to be reassuring and, although this complication can contribute to unwanted outcomes in patients with comorbidities, it rarely causes death on its own [57].

Bone damage is another possible complication of IO drugs administration that raises concerns. However, even if some authors found that bone marrow necrosis occurred more importantly with undiluted sodium bicarbonate IO administration [23], a long-term bone damage assessment animal experiment showed no significant differences in marrow cellularity and cell precursor counts between the undiluted or diluted sodium bicarbonate and the control limb at 3 months [25]. These data are reassuring concerning the long-term impacts on bones with IO drugs administration.

In most of the studies reporting pharmacokinetic parameters, IO drug administration resembles central IV administration more nearly than peripheral IV administration. Medications that normally require central IV access could thus potentially be given through the IO route.

In poisoned patients who are hemodynamically unstable, it may be difficult to obtain a peripheral IV access, and central access requires more time than obtaining an IO access [3]. Although the information retrieved in this review is mostly from non-poisoned cases, the described safety of these drugs in hemodynamically unstable models could be extrapolated to poisoning models. Lack of data for most antidotes, and even more so in human models, makes it important to conduct further studies to determine if administration of antidotes by the IO route is effectively safe and appropriate. The antidotes to be targeted for future studies should be those that need rapid administration and for which no alternative routes to IV exist, during intoxication treatment. Antidotes such as digoxin immune FAB, sodium bicarbonate, calcium chloride, lipid emulsion, insulin, fomepizole, and hydroxocoablamin are a few examples. Although the IM route is possible for atropine and pralidoxime, it is associated with a longer onset of action compared with the IV route and perhaps the IO route offers a faster systemic delivery for these antidotes as well.

Limitations

This study suffers from the usual limitations applicable to systematic reviews in that the data presented are contingent on published results. We performed a thorough search with no limitations of language, and included conference abstracts if they contained all the data required for this review. We also assessed experiments on animals if their model was applicable to human overdose scenarios. Unfortunately, only a small number of antidotes have reports of their use with the IO route published and thus the analysis is not inclusive of all the antidotes a clinician might want to use in practice.

The human evidence is limited to case reports and case series, which by definition are of low GRADE. The collection of studies is also heterogeneous and results cannot be pooled. Another limitation is that much of the data arise from healthy animals, and few experiments used a model of poisoning that would allow better correlation with human poisoning clinical scenarios.

Conclusion

Despite the low-quality evidence available and the absence of human data for many antidotes, IO access seems to be an option for antidotal treatments in toxicological resuscitation if IV access is unavailable. Further research on IO administration of antidotes should focus especially on common antidotes such as acetylcysteine, calcium gluconate, digoxin immune FAB, and fomepizole, as well as antidotes needed for critically ill poisoned patients.

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Appendix 1 – result details

AUC: area under the curve; BP: blood pressure; bpm: beat per minute; CaCl: calcium chloride; C_{max}: peak plasma concentration; HR: heart rate; ICU: intensive care unit; IM: intramuscular; IO: intraosseous; IV: intravenous; MAP: mean arterial pressure; NS: normal saline; SBP: systolic blood pressure; SVR: systemic vascular resistance; T1/2: half time of elimination; T_{max}: time to peak plasma concentration.

Antidote	Type of study	Source	Clinical context	Intervention	Subjects	Outcomes	Methodological quality
Atropine	Animal studies	Murray et al. [21]	Healthy animals	Sequential administration Group 1 (IO): Atropine 0.12 mg/kg Pralidoxime 25 mg/kg Hydroxocobalamin 75 mg/kg Group 2 (IV central line): Atropine 0.12 mg/kg Pralidoxime 25 mg/kg Hydroxocobalamin 75 mg/kg Group 3 (IM): Atropine 0.12 mg/kg Pralidoxime 25 mg/kg *IO proximal tibia	Male Göttingen minipigs randomized into three groups to assess antidotes bioavailability given by different administration route (4–8 minipigs per group)	T_{max} was similar for groups 1 and 2, longer for group 3. T_{max}: Group 1: Atropine: 2 min, Pralidoxime: 1.75 min, Hydroxocobalamin: 10.75 min Group 2: Atropine: 2 min, Pralidoxime: 1.75 min, Hydroxocobalamin: 7.75 min Group 3: Atropine: 3.5 min, Pralidoxime: 8 min For atropine and pralidoxime, group 1 produced higher plasma concentration than group 2 C_{max}: Group 1: Atropine: 117.1 ng/mL, Pralidoxime: 94.7 µg/mL, Hydroxocobalamin: 0.41 mg/mL Group 2: Atropine: 80.9 ng/mL, Pralidoxime: 84.2 µg/mL, Hydroxocobalamin: 0.43 mg/mL Group 3: Atropine: 33.6 ng/mL, Pralidoxime: 35.8 µg /mL The mean AUC-time curve by the IO route was similar to the IV route for all 3 drugs and similar to the IM route for atropine and pralidoxime AUC: Group 1: Atropine: 1625, Pralidoxime: 2759, Hydroxocobalamin: 56.2 Group 2: Atropine: 1236, Pralidoxime: 2206, Hydroxocobalamin: 56.4 Group 3: Atropine: 1631, Pralidoxime: 2173 (Units for AUC: atropine: min ng/mL, Pralidoxime: min µg /mL, Hydroxocobalamin: min mg/mL)	ARRIVE: 18/20 *Study not statistically powered to assess differences between the three groups*
		Orlowski et al. [23]	Healthy animals	Group 1: IO control normal saline Group 2: IO Epinephrine 0.01 mg/kg Group 3: IO Epinephrine in shock dogs Group 4: IO Sodium Bicarbonate 1 mEq/kg (undiluted) Group 5: IO 6% Hydroxyethyl starch in NS 10 mL/kg Group 6: IO Dextrose 50% in water 0.25 g/kg Group 7: IO Lidocaine 1 mg/kg Group 8: IO Calcium Chloride 10 mg/kg Group 9: IO Epinephrine + Sodium Bicarbonate + **Atropine 0.01 mg/kg**	30 healthy dogs divided into 10 groups ($n = 3$) to assess risk of fat and bone marrow emboli to the lungs	Fat and bone marrow emboli to the lungs found in all animals. No significant difference in mean number of emboli/mm^2 of lung among the emergency drugs or solutions and compared with controls. No significant alteration in PaO_2 during the 4-h study period Group 4: Amount of emboli fewest than all groups Groups 7 and 8: Most emboli of the single-drug therapies (groups 1–6) Group 10: Amount of emboli greatest than all groups Examination of femur showed: Group 4 and 10 (sodium bicarbonate) greatest degree of bone marrow necrosis and hemorrhage. The second greatest bone marrow damage, group 6 then group 7	ARRIVE: 15/20

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Antidote	Type of study	Source	Clinical context	Intervention	Subjects	Outcomes	Methodological quality
				Group 10: IO Epinephrine + Sodium Bicarbonate *IO distal femur			
		Prete et al. [26]	Healthy animals	Route 1: IO Atropine 0.01%, 0.03 mg/kg (proximal tibia) ($n = 5$) Route 2: IV Atropine 0.01%, 0.03 mg/kg (saphenous vein) ($n = 5$) Route 3: Endotracheal Atropine 0.01%, 0.03 mg/kg ($n = 6$) *All animals received atropine via IV, endotracheal and IO route with interval of 7 days.	6 healthy adult pigtail macaques in a triple crossover design to compare IV, endotracheal and IO route	Mean Tmax was shorter with IV (1.37 min) than IO (3.87 min) and faster with IO than endotracheal (9.62 min). The mean IV atropine plasma concentration was significantly higher than the IO atropine plasma concentration at 1.25 min. AUC estimates from mean data was higher with IO route (12.57 nmoles/hr/L) compared with IV route (9.83 nmoles/hr/L) and endotracheal route (5.54 nmoles/hr/L).	ARRIVE: 13/20
		Yost et al. [33]	Normovolemic & hypovolemic animals	Group 1: normovolemic, IV atropine ($n = 6$) Group 2: normovolemic, IM atropine ($n = 6$) Group 3: normovolemic, IO atropine ($n = 6$) Group 4: hypovolemic, IV atropine ($n = 5$, 1 died) Group 5: hypovolemic, IM atropine ($n = 6$) Group 6: hypovolemic, IO atropine (n = 6) *For all groups: atropine 2 mg (fixed dose) *IO proximal tibia *IV ear vein *IM gluteal muscle	36 healthy Yorkshire-cross swine randomly assigned to receive atropine via IV, IM or IO routes in normo or hypovolemic state to assess pharmacokinetic parameters	IV and IO groups in both the normovolemic and hypovolemic models reached C_{max} immediately (T_{max}: 0 min) and had a very rapid distribution phase. C_{max} was significantly lower for the IM groups. T_{max} was significantly longer for the IM groups. Normovolemic swine: C_{max}: IV: 374 ± 166 ng/mL, IO: 300 ± 250 ng/mL and IM: 20 ± 8 ng/mL T_{max}: IV: 0 min, IO: 0 min, IM: 6.6 ± 3.1 min AUC: IV: 912 ± 334, IO: 1023 ± 384, IM: 937 ± 183 Hypovolemic swine: C_{max}: IV: 513 ± 221 ng/mL, IO: 309 ± 168 ng/mL and IM: 14 ± 5 ng/mL T_{max}: IV: 0 min, IO: 0 min, IM: 19.5 ± 9.8 min AUC: IV: 1599 ± 392, IO: 1470 ± 523, IM: 1247 ± 168 AUC units: min ng/mL There was a significant increase in absorption time with IM administration in the hypovolemic model compared to the normovolemic model.	ARRIVE= 17/20
	Human cases	Frascone et al. [37]	Burn and Cardiac Arrest	IO Atropine 1 mg push *IO sternum	54 y.o. woman found at a house fire with 65% of burned body surface in cardiac arrest	- Mortality: Patient died 6 hours after arrival	GRADE VERY LOW
		Iserson [43]	Cardiac arrest	IO administration of sodium bicarbonate, Lidocaine, **Atropine,** and vasopressors *IO medial malleolus *No information on doses	22 patients who arrived in ED in cardiac arrest in whom IV line was not established and had an IO needle inserted	- Mortality: 20 patients died in the ED, two others in the hospital within 24 h - Hemodynamics: in two patients, bradycardia responded to atropine within 1 min	Grade very low

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Antidote	Type of study	Source	Clinical context	Intervention	Subjects	Outcomes	Methodological quality
						- Relevant labs: Increase in pH in four of seven patients who received IO sodium bicarbonate. No possible evaluation for the three others - Safety: No evidence of complications from IO access	
		Joseph and Tobias [44]	Anesthetic care in operating room	IO Atropine 5 µg /kg *IO right tibia	1 Infant undergoing anesthesia	- Mortality: Infant survived the surgery - Adverse Effects: 8 months follow-up no leg length discrepancy - Other IO medication: Ampicillin 50 mg/kg, Propofol 1 mg/kg, Vancomycin 10 mg/kg	GRADE VERY LOW
		Manley [46]	Poisoning	IO atropine sulfate (large dose administered to maintain a pulse rate of 100 bpm, no specific dose mentioned) *IO right anteromedial proximal tibia	2-year-old child intoxicated with liquid flea-killer malathion with seizures	- Mortality: child transferred to tertiary pediatric care center, then discharged after 5 days with no residual sequelae - Adverse effects: No complications were reported in subsequent follow-up examinations - Also received by IO route: Pancuronium bromide to control seizure activity refractory to diazepam and phenobarbital	GRADE VERY LOW
		McNamara et al. [48]	Burn and cardiac arrest	IO Atropine and sodium bicarbonate *No information on IO localization and on medication doses	Two infants (5 and 17 months old) in cardiac arrest found at a scene of house fire	- Mortality: both children died	Grade very low
		Ramet et al. [52]	Shock	IO Sodium bicarbonate IO Atropine *No information about doses *IO tibial plate	1800 g preterm neonate	- Mortality: infant died - Relevant labs: improvement of blood gases and electrolytes (no values) - Adverse effects: at autopsy no histological abnormalities at insertion site and bone marrow space - Other IO medication: human albumin 10 mL/kg $\times$2, Dextrose 10% 130 mL/kg/d, epinephrine, vancomycin, cefotaxime, dobutamine, and fentanyl	Grade very low
		Sacchetti et al. [53]	Cardiac Arrest	IO Atropine 0.1 mg push *IO proximal tibia	1 infant in cardiac arrest	- Mortality: Infant survived - Hemodynamics: Following administration of IO epinephrine 0.1 mg, Atropine 0.1 mg and 2nd dose of epinephrine: HR went from asystole to 168 bpm with SBP: 90 mmHg	GRADE VERY LOW
Calcium chloride	Animal studies	Orlowski et al. [23]	Healthy animals	Group 1: IO Control: NS Group 2: IO Epinephrine 0.01 mg/kg Group 3: IO Epinephrine in shock dogs Group 4: IO Sodium Bicarbonate 1 mEq/kg (undiluted) Group 5: IO 6% Hydroxyethyl starch in NS 10 mL/kg Group 6: IO Dextrose 50% in water 0.25 g/kg	30 healthy dogs divided into 10 groups (n:3) to assess risk of fat and bone marrow emboli to the lungs	Fat and bone marrow emboli to the lungs found in all animals. No significant difference in mean number of emboli/mm^2 of lung among the emergency drugs or solutions and compared with controls. No significant alteration in PaO_2 during the 4-hour study period. Group 4: Amount of emboli fewest than all groups Group 7 and 8: Most emboli of the single-drug therapies (groups 1 to 6)	ARRIVE: 15/20

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Antidote	Type of study	Source	Clinical context	Intervention	Subjects	Outcomes	Methodological quality
				Group 7: IO Lidocaine 1 mg/kg Group 8: IO **Calcium Chloride** 10 mg/kg Group 9: IO Epinephrine + Sodium Bicarbonate + Atropine 0.01 mg/kg Group 10: IO Epinephrine + Sodium Bicarbonate *IO distal femur		Group 10: Amount of emboli greatest than all groups. Examination of femur showed: Group 4 and 10 (Sodium Bicarbonate) greatest degree of bone marrow necrosis and hemorrhage. The second greatest bone marrow damage, group 6 then group 7.	
		Orlowski et al. [24]	Healthy animals	Group 1: Epinephrine 0.01 mg/kg in a normotensive model Group 2: Sodium Bicarbonate 1 mEq/kg **Group 3: CaCl 10 mg/kg** Group 4: Hydroxyethyl starch 10 mL/kg Group 5: Dextrose 50% in water 250 mg/kg Group 6: Lidocaine 1 mg/kg Group 7: Epinephrine 0,01 mg/kg in a shock model *All animals were treated with 3 routes: IO (distal femur), central and peripheral IV	21 healthy dogs randomised in 7 groups treated with specific emergency drugs by 3 different routes of administration	Change in ionized calcium levels over time after CaCl bolus was not statistically different between the 3 routes C_{max} for calcium was 0.8 mmol/L for IO route, 1.2 mmol/L for central IV and 1.1 mmol/L for peripheral IV (NS) *CaCl concentration from graph T_{max} was 15 seconds for the 3 routes of administration.	ARRIVE:14/20
Dextrose	Animal studies	Neish et al. [22]	Healthy animals	Group 1: IO Dextrose 50% 1 mL/kg (0.5 g/kg) (n:10) Group 2: IV Dextrose 50% 1 mL/kg (0.5 g/kg) (n:4) *IO: sternum (n:7) and iliac crest (n:3) *IV: jugular vein	14 healthy domestic swine divided in two groups to compare IO D50% effect vs IV D50% effect.	Mortality: All swine survived Relevant labs: Mean blood glucose level results at each time interval were similar between group 1 and 2. Group 1: At 1 min: 18.5 mmol/L, 3 min: 14.8 mmol/L, 5 min: 13 mmol/L, 10 min: 10.2 mmol/L, 15 min: 9.2 mmol/L Group 2: At 1 min: 19.4 mmol/L, 3 min: 15.7 mmol/L, 5 min: 13.45 mmol/L, 10 min: 11.55 mmol/L, 15 min: 10.65 mmol/L *BG values from graph Baseline blood glucose level was statistically higher in group 1 (7.4 ± 3.3 mmol/L) than in group 2 (4.8 ± 0.2 mmol/L). T_{max} for glucose was approximately 1 min for group 1 and 2. *Value from graph	ARRIVE: 14/20
		Orlowski et al. [23]	Healthy animals	Group 1: IO Control: NS Group 2: IO Epinephrine 0.01 mg/kg	30 healthy dogs divided into 10 groups ($n = 3$) to assess risk of fat and bone	Fat and bone marrow emboli to the lungs found in all animals. No significant difference in mean number	ARRIVE: 15/20

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Antidote	Type of study	Source	Clinical context	Intervention	Subjects	Outcomes	Methodological quality
				Group 3: IO Epinephrine in shock dogs Group 4: IO Sodium Bicarbonate 1 mEq/kg (undiluted) Group 5: IO 6% Hydroxyethyl starch in NS 10 mL/kg **Group 6: IO Dextrose 50% in water 0.25 g/kg** Group 7: IO Lidocaine 1 mg/kg Group 8: IO CaCl 10 mg/kg Group 9: IO Epinephrine + Sodium Bicarbonate + Atropine 0.01 mg/kg Group 10: IO Epinephrine + **Sodium Bicarbonate** *IO distal femur	marrow emboli to the lungs	of emboli/mm^2 of lung among the emergency drugs or solutions and compared with controls. No significant alteration in PaO_2 during the 4-hour study period. Group 4: Fewest amount of emboli Group 7 and 8: Most emboli of the single-drug therapies (groups 1 to 8) Group 10: Amount of emboli greatest than all groups. Examination of femur: Group 4 and 10 (Sodium Bicarbonate) greatest degree of bone marrow necrosis and hemorrhage. The second greatest bone marrow damage, group 6, then group 7.	
		Orlowski et al. [24]	Healthy animals	Group 1: Epinephrine 0.01 mg/kg in a normotensive model Group 2: Sodium Bicarbonate 1 mEq/kg Group 3: Calcium Chloride 10 mg/kg Group 4: Hydroxyethyl starch 10 mL/kg Group 5: Dextrose 50% in water 250 mg/kg Group 6: Lidocaine 1 mg/kg Group 7: Epinephrine 0.01 mg/kg in a shock model *Each animal in each group treated with all three routes: IO (distal femur), central IV, peripheral IV	21 healthy dogs randomised in 7 groups treated with specific emergency drugs by 3 different routes of administration	C_{max} glucose levels after 50% dextrose administration was similar for IO (34.4 mmol/L) and central IV (31.1 mmol/L) (NS). Peripheral IV route had a statistically lower peak concentration of 23.3 mmol/L. *BG values from graph Time to reach the peak concentration was 15 seconds for the 3 routes.	ARRIVE: 14/20
	Human cases	Mortensen et al. [50]	Poisoning	IO Sodium bicarbonate 2.6 mEq/kg IO Dextrose 50% (unrecorded volume) *IO proximal tibia	6-month-old infant intoxicated with 25 mg of encainide	- Mortality: infant survived - Hemodynamics: impalpable femoral pulse to palpable pulse and BP 108/63 mmHg - Electrocardiographic variables: wide QRS (0.16 s) and ventricular fibrillation to sinus rhythm with normalization of QRS - Mental status: recovery of consciousness - Other relevant information: Received IO phenytoin 20 mg/kg, fluids. CPR cardioversion and defibrillation were performed	Grade very low

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Antidote	Type of study	Source	Clinical context	Intervention	Subjects	Outcomes	Methodological quality
		Nafiu et al. [51]	Anaphylaxis Cardiac arrest	IO Sodium bicarbonate 8.4% 2 mL/kg diluted IO Dextrose 50% 1 mL/kg (0.5 g/kg) diluted *IO proximal tibia	One infant in cardiac arrest following anaphylaxis	- Mortality: infant survived - Hemodynamics: return to cardiac activity after first of five doses of IO epinephrine - Other IO medication: hydrocortisone 5 mg/kg, aminophylline 5 mg/kg and NS 50 mL.	Grade very low
Diazepam	Animal studies	Beall et al. [10] [abstract]	Seizure	Group 1: IV Diazepam 4 mg ($n = 7$) Group 2: IO Diazepam 4 mg (IO manually installed) ($n = 8$) Group 3: IO Diazepam 4 mg (prototype automated IO device) ($n = 7$) *IO upper tibia *IV site not mentioned	22 Yorkshire piglets with induced seizure divided in 3 groups to receive diazepam via IV injection, IO (manually place IO needle) or IO (prototype automated IO device) *In each group some animals served as control and received saline instead of pentylenetetrazol.	Mental status: The 3 delivery methods resulted in temporary cessation of EEG activity in the seizing piglets (no specific time mentioned) Relevant labs: T_{max} of 1–3 min post diazepam for the 3 routes C_{max} were not statistically different among the 3 groups. (group 1: 908 + 220 ng/mL; group 2: 871 + 536 ng/mL; group 3: 675 + 465 ng/mL) (NS)	ARRIVE: 4/20
		Brickman et al. [15]	Healthy animals	Group 1: IO Diazepam 3 mg/kg x1 ($n = 10$) Group 2: IV Diazepam 3 mg/kg x 1 ($n = 10$) *IO proximal tibia *IV peripheral vein	20 healthy mongrel dogs randomized into 2 equal groups to compare IO versus IV diazepam infusion	Group 1 mean diazepam levels at each time interval were slightly less than group 2 but the difference was not statistically significant. Group 1: at 1 min: 1740 ng/mL 3 min: 710 ng/mL, 6 min: 400 ng/mL, 10 min: 360 ng/mL, 20 min: 210 ng/mL Group 2: at 1 min: 2040 ng/mL, 3 min: 1200 ng/mL, 6 min: 740 ng/mL, 10 min: 560 ng/mL, 20 min: 370 ng/mL (NS)	ARRIVE: 14/20
		Spivey et al. [29]	Seizure	Group 1: IO Diazepam 0.1 mg/kg Group 2: IV Diazepam 0.1 mg/kg Group 3: Control (no drug administered) *IO proximal tibia *IV peripheral vein	15 domestic swine divided into 3 equal groups ($n = 5$) to assess IO diazepam in induced seizure model swine. Pentylenetetrazol 100 mg/kg IV to induce seizure	Mortality: All swine survived Mental status: time to control epileptogenic activity did not show a significant difference between IV or IO. However, time to complete suppression of epileptogenic activity: IV: 2 min IO: 8 min Relevant labs: The mean diazepam concentration between IV and IO over time. Group 1: at 1 min: 187.5 ng/mL, 2 min: 175 ng/mL, 5 min: 150 ng/mL, 10 min: 150 ng/mL, 15 min: 143.75 ng/mL, 20 min: 143.75 ng/mL (NS). Group 2: at 1 min: 250 ng/mL, 2 min: 200 ng/mL, 5 min: 187.5 ng/mL, 10 min: 125 ng/mL, 15 min: 131.25 ng/mL, 20 min: 125 ng/mL (NS) *Values from graph Group 2 had higher diazepam level during the first 5 min and group 1 had higher diazepam level for the remaining 15 min.	ARRIVE: 15/20

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Antidote	Type of study	Source	Clinical context	Intervention	Subjects	Outcomes	Methodological quality
	Human cases	McNamara et al. [48]	Seizure	IO Diazepam 1 mg push (estimated infant weight 8 kg: 0.125 mg/kg) *IO proximal tibia	1 infant with seizures	- Neurological outcomes: Seizures stopped within one min of diazepam	GRADE VERY LOW
Hydroxocobalamin	Animal studies	Bebarta et al. [12]	Poisoning	Group 1: IO hydroxocobalamin 150 mg/kg × 1 in a total of 180 mL ($n = 12$) Group 2: IV hydroxocobalamin 150 mg/kg × 1 in a total of 180 mL ($n = 12$) *IO proximal tibia *IV femoral vein	24 swine randomized to 2 groups, intoxicated with IV cyanide 4% until severe hypotension (50% of baseline Mean arterial pressure (MAP))	- Mortality: 2/12 in each group - Hemodynamics: Group 1-2 had a similar return to baseline MAP from hypotension. No significant differences for cardiac output, SvO2 and systemic vascular resistance - Adverse effects: no difference in complications related to infusion between the groups. No concerning necrosis or tissue damage observed at IO sites by pathologist in 5 animals.	ARRIVE: 18/20
		Bebarta et al. [11]	Hemorrhagic shock	Group 1: IV 150 mg/kg in hydroxocobalamin in saline to a total of 180 mL ($n = 10$) Group 2: No treatment (negative control group, $n = 10$) Group 3: IO 500 mL whole blood (positive control group, $n = 8$) Group 4: IO 150 mg/kg in hydroxocobalamin in 180 mL of saline ($n = 8$)	36 swine in class III hemorrhagic shock (30% of blood volume removed), randomized to 4 groups to assess if IO hydroxocobalamin is comparable to other treatments over no treatment on hemodynamics for 60 min.	Mortality: 1/8 in group 3 Hemodynamics: At 60 min mean SBP was significantly greater in the treatment groups (groups 1, 3 and 4) than in the control group (group 2) ($p < 0.01$) Group 1: 75.1 (SD± 7.5) mmHg Group 2: 55.3 (SD ±12.9) mmHg Group 3: 78.25 (SD ±15.8) mmHg Group 4: 83.7 (SD ±17.3) mmHg MAP significantly improved over time in treated groups but not in the control group. Values on a graph but not specifically reported. At 10 min post treatment, systemic vascular resistance (SVR) was higher in group 4 (IO hydroxocobalamin) compared to group 1 (IV hydroxocobalamin) and group 3 (IO whole blood). At 20 min and until the end (60 min) SVR was not significantly different between the 3 treatment groups (group 1-3-4) HR was not different among the 3 treated groups (groups 1-3-4) during the whole experiment No difference over time between the 4 groups for cardiac output. Serum lactate was improving in all treated groups (groups 1-3-4); it was deteriorating in the non-treated group (group 2)	ARRIVE: 18/20 (no for allocating animals to experimental groups and no for adverse events)
		Borron et al. [14]	Healthy animals	Group 1: IO Hydroxocobalamin 75 mg/kg (3 mL/kg) over 7.5 min ($n = 6$) Group 2: IO NS 3 mL/kg over 7.5 min ($n = 6$)	16 healthy Spanish goats randomized into 3 groups to assess hemodynamics after hydroxocobalamin administration	Mortality: All goats survived and were sacrificed at the end of experiment Hemodynamics: Group 1: mean increases of SBP is 14% (peak at 120 min) and DBP is 17% (peak at 120 min)	ARRIVE: 19/20

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Antidote	Type of study	Source	Clinical context	Intervention	Subjects	Outcomes	Methodological quality
				Group 3: Practice group ($n = 4$) *IO 4 anatomical locations: right or left forelimb, right or left hind limb		Group 2: mean increases of SBP is 36% (peak at 60 min) and DBP is 42% (peak at 110 min) No statistical difference in the mean values of SBP, DBP and HR over the time among group 1-2. Adverse effects: Well tolerated	
		Lairet et al. [20]	Hemorrhagic shock	Group 1: IO Hydroxocobalamin 150 mg/kg Group 2: IO Hydroxocobalamin 70 mg/kg Group 3: IO Hydroxocobalamin 35 mg/kg *IO in proximal tibia *Drug infused over 7 min	6 swine in hemorrhagic shock divided (blood removed until 50% of baseline MAP) in 3 groups ($n = 2$) to assess IO hydroxocobalamin in improving blood pressure	- Mortality: All swine survived - Hemodynamics: Significant difference in mean arterial blood pressure between group 1 to group 2 and 3 ($p < 0.001$) (values not reported). The mean systematic vascular resistance at 90 min post treatment was: Group 1: 1025 (SD 194) Group 2: 1099 (SD 135) Group 3: 726 (SD 130) Hydroxocobalamin significantly increased mean arterial blood pressure and systemic vascular resistance. The blood pressure response paralleled increasing drug dose. - Adverse effects: missing data	ARRIVE: 11.5/20
		Murray et al. [21]	Healthy animals	Sequential administration Group 1 (IO): Atropine 0.12 mg/kg Pralidoxime 25 mg/kg Hydroxocobalamin 75 mg/kg Group 2 (IV central line): Atropine 0.12 mg/kg Pralidoxime 25 mg/kg Hydroxocobalamin 75 mg/kg Group 3 (IM): Atropine 0.12 mg/kg Pralidoxime 25 mg/kg *IO proximal tibia	Male Göttingen minipigs randomized into 3 groups to assess antidotes bioavailability given by different administration route (4 to 8 minipigs per group)	T_{max} was similar for group 1-2, longer for group 3. T_{max}: Group 1: Atropine: 2 min, Pralidoxime: 1.75 min, Hydroxocobalamin: 10.75 min Group 2: Atropine: 2 min, Pralidoxime: 1.75 min, Hydroxocobalamin: 7.75 min Group 3: Atropine: 3.5 min, Pralidoxime: 8 min For atropine and pralidoxime, group 1 produced higher plasma concentration than group 2. C_{max}: Group 1: Atropine: 117.1 ng/mL, Pralidoxime: 94.7 µg /mL, Hydroxocobalamin: 0.41 mg/mL Group 2: Atropine: 80.9 ng/mL, Pralidoxime: 84.2 µg /mL, Hydroxocobalamin: 0.43 mg/mL Group 3: Atropine: 33.6 ng/mL, Pralidoxime: 35.8 µg /mL However, the mean area under the concentration-time curve by the intraosseous route was similar to the intravenous route for all 3 drugs and similar to the intramuscular route for atropine and pralidoxime. AUC: Group 1: Atropine: 1.625, Pralidoxime: 2.759, Hydroxocobalamin: 56.2 Group 2: Atropine: 1.236, Pralidoxime: 2.206, Hydroxocobalamin: 56.4 Group 3: Atropine: 1.631, Pralidoxime: 2.173	ARRIVE: 18/20 *Study not statistically powered to assess differences between the 3 groups.*

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Antidote	Type of study	Source	Clinical context	Intervention	Subjects	Outcomes	Methodological quality
	Human cases	Fortin et al. [36]	Poisoning	IO Hydroxocobalamin 2.5 g prehospital administration *IO site not mentioned	9-month-old infant with suspected smoke inhalation cyanide poisoning	- Mortality: Infant survived - Hemodynamics: parameters stabilized after antidote (before treatment: HR 200 bpm, BP 120/70 mmHg and oxygen saturation 92%) - Neurological status: Glasgow Coma Scale score 8 before treatment, no neurological sequelae	GRADE VERY LOW
Insulin	Human cases	Alawi et al. [34]	Pediatric dehydration and DKA	IO Infusion of Insulin started at 0.1 unit/kg/h and decreased to 0.08 unit/kg/h at 10 hours for duration of 14 hours *IO proximal tibia	1 infant with severe diabetic ketoacidosis and dehydration	- Mortality: Infant survived - Relevant labs: Correction of hyperglycemia and metabolic acidosis (BG graphic only values) - Safety: no complication associated with the IO line was detected	GRADE VERY LOW
		Gallo et al. [39]	Pediatric DKA	IO NS 0.9% and insulin therapy for 12 hours. Doses not mentioned *IO right tibial plate	18-month-old and 3-year-old children - comatose, arterial sample revealing DKA	- Mortality: Infants survived - Safety: no treatment complications Other relevant information: 12 hours later IV access established and IO removed.	GRADE VERY LOW
Lipid emulsion	Animal studies	Fettiplace et al. [17]	Poisoning	Group 1: IO 10 mL/kg Lipid emulsion 30% ($n = 6$) Group 2: IO 10 mL/kg Saline ($n = 6$) Group 3: IV 10 mL/kg Lipid emulsion 30% ($n = 6$) Group 4: No treatment ($n = 7$) Antidotes administered 10 seconds after bupivacaine infusion *IO proximal femur	25 Sprague-Dawley male rats, intoxicated with IV bupivacaine (10 mg/kg over 20 seconds) randomized to 4 groups to assess hemodynamics response after IO or IV Lipid emulsion.	- Mortality: All rats survived - Hemodynamics: Group 1-3 similar rate of hemodynamics recovery (return to 50% of baseline carotid blood flow) Group 1 significantly recovered faster than Group 2 or Group 4. After 480 seconds group 1-3 both recovered to baseline carotid blood flow. All groups had persistent decreased HR coupled with an increase in SBP and DBP for group 1-3 - Adverse effects: missing data	ARRIVE: 12/20
	Human cases	French et al. [38]	Poisoning	IO Lipid emulsion (20% solution) 12 mL bolus then an infusion *No information on infusion rate *IO site not mentioned	1 infant with tonic-clonic seizures secondary to lidocaine toxicity following iatrogenic IO overdose	- Mortality: Patient recovered uneventfully - Neurological status: No further seizures - Other relevant information: toxic dose: 10 mL of 1% lidocaine (10.1 mg/kg)	GRADE VERY LOW
		Sampson and Bedy [55]	Poisoning	IO Intravenous Lipid emulsion (IFE) 20% * IO proximal tibia	24-year-old female following massive deliberate verapamil overdose (30 tablets of 240-mg extended-release verapamil along with a smaller but unknown quantity of 80 mg immediate-release verapamil. The ingestion occurred 1 to 2 hours before her arrival.)	- Mortality: patient transferred to ICU, died 2 days later - Other relevant information: initial treatment of glucagon, high dose insulin (100 units), and calcium gluconate with no change in BP, norepinephrine drip via central line (up to 20 mcg per min). Half way during IO administration infusion not flowing & IFE switched to IV site.	GRADE VERY LOW
Methylene blue	Animal studies	Hosseinpour and Khodaiari [19]	Healthy animals	Group 1: IO 10 mL (10 mg) of methylene blue at 2.5 mL/5 seconds ($n = 10$) Group 2: IV 10 mL (10 mg) of methylene blue at	20 white male adult New Zealand rabbits divided into 2 equal groups to compare appearance time of methylene blue in the aorta.	No significant differences between the appearance of methylene blue into the aorta between 2 groups.	ARRIVE: 16/20

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Antidote	Type of study	Source	Clinical context	Intervention	Subjects	Outcomes	Methodological quality
				2.5 mL/5 seconds into the marginal vein of either ear ($n = 10$) *IO proximal left tibia			
	Human cases	Herman et al. [42]	Poisoning + Pediatric dehydration	IO Methylene Blue 1% 1 mg/kg (0.3 mL) over 3-5 min IO Sodium Bicarbonate 2 mEq/kg *IO proximal tibia	6-week-old infant with acquired methemoglobinemia	- Mortality: Infant survived - Oximetry rose from 86% to 98-100% 8 min after antidote administration and cyanosis resolved - Relevant labs: Methemoglobin concentration decreased from 29.3% to 8.2% (3 hours after IO antidote administration) and 4.4% at 18.3 hours (received a 2nd methylene blue dose via IV route at 7.8 hours), acidosis at arrival (pH 7.21 before treatment). - Safety: No apparent side effect from the IO drugs administration.	GRADE VERY LOW
Naloxone	Human cases	Lameijer et al. [45]	Poisoning	IO Naloxone 0.4 mg push x 2 given by paramedics *IO site not mentioned	1 multi-drug user adult of 44 y.o. with suspected methadone overdose	- Mortality: patient's condition stabilized, but required ICU admission - Hemodynamics: tachypnea, respiratory alkalosis, tachycardia, then a distributive or cardiac shock - Neurological outcomes: reduced Glasgow coma score and progressive agitation leading to medical sedation and intubation - Electrocardiographic variables and adverse effect: ventricular tachycardia 160-180 bpm (Naranjo scale possible to probable) treated first with magnesium sulfate and amiodarone, then with an electro cardioversion after a sudden hypotension - Other relevant information: The patient had multiple factors that could make him more prone to dysrythmogenic events (congenital heart disease, chronically high opiate doses and multi-drug user).	GRADE VERY LOW
Phentolamine	Human cases	Greenstein et al. [41]	Shock and Extravasation	IO Dopamine infusion at 10 µg/kg/min and titrated for goal mean arterial pressure greater than or equal to 65 mmHg IO norepinephrine infusion at 0.5 µg/kg/min added while dopamine titrated off IO Phentolamine 5 mg push into the IO needle *IO right lower proximal tibia	A 42-year-old man hospitalized for intestinal pseudo-obstruction, in shock treated with dopamine and norepinephrine through IO route, who underwent vasopressors extravasation into the soft tissues and threatened limb perfusion related to intraosseous access use	- Mortality: Patient survived - Safety: salvage of limb after local nitroglycerin 2% ointment, IO phentolamine and injection of vasodilators into the right common femoral artery (verapamil 2.5 mg and Nitroglycerin 0.2 mg) - Hemodynamics: return of palpable dorsalis pedis and posterior tibialis pulses with color flow Doppler throughout the common femoral artery, superficial artery and both pedal arteries. Disappearance of limb mottledness within a few min after treatments.	GRADE VERY LOW

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Antidote	Type of study	Source	Clinical context	Intervention	Subjects	Outcomes	Methodological quality
Pralidoxime	Animal studies	Murray et al. [21]	Healthy animals	Sequential administration Group 1 (IO): Atropine 0.12 mg/kg Pralidoxime 25 mg/kg Hydroxocobalamin 75 mg/kg Group 2 (IV central line): Atropine 0.12 mg/kg Pralidoxime 25 mg/kg Hydroxocobalamin 75 mg/kg Group 3 (IM): Atropine 0.12 mg/kg Pralidoxime 25 mg/kg *IO proximal tibia	Male Göttingen minipigs randomized into 3 groups to assess antidotes bioavailability given by different administration route (4 to 8 minipigs per group)	T_{max} was similar for group 1-2, longer for group 3. T_{max}: Group 1: Atropine: 2 min, Pralidoxime: 1.75 min, Hydroxocobalamin: 10.75 min. Group 2: Atropine: 2 min, Pralidoxime: 1.75 min, Hydroxocobalamin: 7.75 min Group 3: Atropine: 3.5 min, Pralidoxime: 8 min For atropine and pralidoxime, group 1 produced higher plasma concentration than group 2. C_{max}: Group 1: Atropine: 117.1 ng/mL, Pralidoxime: 94.7 µg /mL, Hydroxocobalamin: 0.41 mg/mL Group 2: Atropine: 80.9 ng/mL, Pralidoxime: 84.2 µg /mL, Hydroxocobalamin: 0.43 mg/mL Group 3: Atropine: 33.6 ng/mL, Pralidoxime: 35.8 µg /mL The mean AUC-time curve by the IO route was similar to the intravenous route for all 3 drugs and similar to the intramuscular route for atropine and pralidoxime. AUC: Group 1: Atropine: 1.625, Pralidoxime: 2.759, Hydroxocobalamin: 56.2 Group 2: Atropine: 1.236, Pralidoxime: 2.206, Hydroxocobalamin: 56.4 Group 3: Atropine: 1.631, Pralidoxime: 2.173	ARRIVE: 18/20
		Uwaydah et al. [30]	Normovolemic & hypovolemic	Group 1: Normovolemic, Pralidoxime Chloride 2 mL, 660 mg (2-PAM Cl) via IM (n:5) Group 2: Normovolemic, 2-PAM Cl 2 mL, 660 mg via IO (n:5) Group 3: Normovolemic, 2-PAM Cl 2 mL, 660 mg via IO, followed by IM 60 min later (n:4) Group 4: Hypovolemic, 2-PAM Cl 2mL, 660 mg via IM (n:4) Group 5: Hypovolemic, 2-PAM Cl 2mL, 660 mg via IO (n:4)	22 immature Yorkshire swine in total 14 normovolemic swine were randomly assigned to 3 groups: IO, IM and IO/IM 2-PAM Cl administration (groups 1, 2 and 3) 8 hypovolemic swine (that bled to a mean arterial pressure of 50 mmHg) were randomly assigned to 2 groups: IO and IM 2-PAM Cl administration (group 4 and 5)	Group 2 significantly reached therapeutic levels faster and had a greater C_{max} compared to group 1. However, the mean transit time (MTT) in plasma was 80% longer in group 1 compared to group 2. Group 1: Time to reach therapeutic level: 2 min, T_{max}: 9.2 min, t1/2: 116 min and C_{max}: 8.5 µg /mL, MTT: 165 min Group 2: Time to reach therapeutic level: 15 seconds, T_{max}: 0 min, t1/2:83 min and C_{max}: 199 µg /mL, MTT: 91 min Group 3 had similar kinetic parameters to group 2 following first IO injection and then similar kinetic parameters to group 1 at 60 min. However, because of the second IM injection at 60 min, 2-PAM concentration never fell below therapeutic levels (4 µg /mL) in group 3 compared to groups 1 and 2.	ARRIVE 18/20

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Antidote	Type of study	Source	Clinical context	Intervention	Subjects	Outcomes	Methodological quality
						Group 3: Time to reach therapeutic level: 15 seconds, T_{max}: 0 min and Initial C_{max}: 229 µg /mL, second peak following IM inj: 10.8 µg /mL Groups 4 and 5 (hypovolemic swine) had longer mean transit time and half-life after either IO or IM infusion when compared to normovolemic swine (group 1 and 2). No statistical differences in terms of 2-PAM plasma concentrations between group 2 (normovolemic IO) and group 5 (hypovolemic IO) (values on graph only) Group 4: Time to reach therapeutic level: 4 min, T_{max}: 8.8 min, t1/2: 275 min and C_{max}: 9.6 µg /mL, MTT: 399 min Group 5: Time to reach therapeutic level: 15 seconds, T_{max}: 0 min, T1/2: 121 min and C_{max}: 325 µg /mL, MTT: 155 min	
Prothrombin complex concentrate	Human cases	Means and Gimbar [49]	NOAC anticoagulation reversal	IO Profilnine SD (factor IX complex [human] [factors II, IX, X]) 1490 U (33 U/kg) reconstituted with 10 mL of manufacturer provided diluent and administered at its maximum rate of 10 mL/min via the IO line. *IO right humerus (IO preferred over IV because of the patient's bruising/ bleeding, Point-of-care Hb, and declining blood pressure)	64 y.o. man presenting to the emergency department with abdominal pain and rectal bleeding. His past medical history includes non–ST elevation myocardial infarction, pacemaker, chronic hyponatremia, adrenal insufficiency, and myositis. -He had been recently prescribed rivaroxaban 15 mg orally twice a day for lower extremity deep vein thrombosis and clopidogrel for recent non–ST elevation myocardial infarction.	Clinical course: patient was transferred to the ICU for further care and US guided IV placed. Melena subsided, hemodynamics stabilized. - Due to other medical conditions, transition to comfort care Relevant labs: not available	GRADE VERY LOW
Sodium bicarbonate	Animal studies	Abdelmoneim et al. [9]	Cardiac arrest	Group 1: control group (no drug) ($n = 6$) Group 2: IO Epinephrine 0.01 mg/kg ×1 dose and 0.1 mg/kg/dose ×2 doses ($n = 6$) Group 3: IO Normal Saline 10 mL/kg/dose ×3 doses ($n = 6$) **Group 4: IO Sodium Bicarbonate** **2 mEq/kg/dose × 3 doses at 5, 10, and 15 min ($n = 6$)** Group 5: IO NS 20 mL/kg, Epinephrine 0.01 mg/kg, then 0.1 mg/kg,	32 mixed-breed piglets in induced anoxic cardiac arrest under prolonged cardiopulmonary resuscitation (30 min) to assess the effect of infusion on acid-base status between IO sites and mixed venous blood in a randomized prospective study	- Mortality: none - Relevant labs: in group 4, pH and PCO_2 of IO blood were higher and clinically different compared with mixed venous site and the contralateral IO site at 10, 15, and 20 min of cardiopulmonary resuscitation. The pH values increased to 7.43 in IO site, 7.28 in contralateral IO site and 7.38 in central venous site at 30 min, while the average pH in the 4 other groups remained between 6.90 and 7.23 in all confounded sites (1 average pH value of 7.37 in central venous site in the control group) In group 5, no statistical difference in pH values between sites, except	ARRIVE 12.5/20

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Antidote	Type of study	Source	Clinical context	Intervention	Subjects	Outcomes	Methodological quality
				Sodium Bicarbonate 2 mEq/kg (sequential adm) ($n = 6$) *IO proximal tibia		after 30 min where central venous site pH was at least 0.1 unit inferior to IO pH values. The pH increased to 7.52 in IO blood 5 min after bicarbonate infusion compared with a pH of 7.07 in mixed venous blood - Adverse effects: not reported	
		Bielski et al. [13]	Healthy animals	Group 1: NS or **sodium bicarbonate 2 mEq/kg (undiluted)** or Dopamine infusion ($n = 10$) Animals killed 24 h after infusion Group 2: Idem to Group 1 ($n = 10$). Animals killed 21 days after infusion.	20 healthy New Zealand immature rabbits divided into 2 equal groups to determine the effects on physis and growth rate of bone at 24 h and 21 d post IO injection	- Mortality: all animals were killed - Safety: none of the treated tibias showed evidence of growth disturbance Equal length between infused tibias and contralateral non-infused tibia Group 1: hematomas confined to puncture site Group 2: normal pattern of trabecular bone in metaphysis	ARRIVE: 12.5/20
		Dedrick et al. [16]	Healthy animals	Group 1: control: no puncture Group 2: IO Puncture only Group 3: IO NS Group 4: IO Sodium Bicarbonate 1 mEq/kg diluted into 1 mL of volume	11 healthy nestling rabbits tibias ($n = 22$) randomized into four groups to assess potential growth plate damage	- Mortality: at 3 or 5 d after infusion animals were killed and tibias removed - Safety: abnormal primary trabeculae in groups 3 and 4 described as newly formed and poorly calcified trabeculae (injured and repair) No histological or radiographic differences between saline and bicarbonate infusions	ARRIVE: 15/20
		Fiallos et al. [18]	Cardiac arrest	Group 1: no IO cannula + CPR ($n = 5$) Group 2: IO cannula with no infusion + CPR ($n = 6$) Group 3: IO epinephrine 0.1 mg/kg push ×3 + CPR ($n = 6$) Group 4: IO NS 10 mL/kg bolus ×3 + CPR ($n = 6$) **Group 5: IO sodium bicarbonate 2 mEq/kg push × 3 + CPR ($n = 8$) (undiluted)** (proximal tibia) Group 6: no IO cannula and no CPR ($n = 2$) *IO proximal tibia	33 mixed breed piglets randomized to six groups of ± IO infusions and ± CPR after induced cardiac arrest to assess incidence and magnitude of fat emboli after CPR and IO infusion	- Mortality: animals who survived after CPR were killed - Relevant labs: pH after 30 min of CPR significantly higher in group 5 (pH 7.51 ± 0.20) compared with the other groups (mean pH 7.34 ± 0.20) - Safety: lung specimens and blood samples analysis: no fat globules in the buffy coat of pulmonary capillary bed. Fat globules in the peribronchial blood vessels and perialveolar region, none in the control group. No difference in appearance, distribution and percentage of fat globules between the five treatment groups	ARRIVE: 15/20
		Orlowski et al. [23]	Healthy animals	Group 1: IO control: NS Group 2: IO Epinephrine 0.01 mg/kg Group 3: IO Epinephrine in shock dogs **Group 4: IO sodium bicarbonate 1 mEq/kg (undiluted)** Group 5: IO 6% Hydroxyethyl starch in NS 10 mL/kg	30 healthy dogs divided into 10 groups ($n = 3$) to assess risk of fat and bone marrow emboli to the lungs	Fat and bone marrow emboli to the lungs found in all animals. No significant difference in mean number of emboli/mm^2 of lung among the emergency drugs or solutions and compared with controls. No significant alteration in PaO_2 during the 4-h study period. Group 4: Fewest amount of emboli Groups 7 and 8: Most emboli of the single-drug therapies (groups 1–6)	ARRIVE: 15/20

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Antidote	Type of study	Source	Clinical context	Intervention	Subjects	Outcomes	Methodological quality
				Group 6: IO Dextrose 50% in water 0.25 g/kg Group 7: IO Lidocaine 1 mg/kg Group 8: IO CaCl 10 mg/kg Group 9: IO Epinephrine + **sodium bicarbonate** + Atropine 0.01 mg/kg Group 10: IO Epinephrine + **sodium bicarbonate** *IO distal femur		Group 10: amount of emboli greatest than all groups Examination of femur: Groups 4 and 10 (sodium bicarbonate) greatest degree of bone marrow necrosis and hemorrhage. The second greatest bone marrow damage, group 6, then group 7	
		Orlowski et al. [24]	Healthy animals	Group 1: Epinephrine 0.01 mg/kg in a normotensive model **Group 2: sodium bicarbonate 1 mEq/kg** Group 3: calcium chloride 10 mg/kg Group 4: hydroxyethyl starch 10 mL/kg Group 5: dextrose 50% in water 250 mg/kg Group 6: lidocaine 1 mg/kg Group 7: epinephrine 0.01 mg/kg in a shock model *Each animal in each group treated with all three routes: IO (distal femur), central IV, peripheral IV	21 dogs randomized in seven groups treated with specific emergency drugs by three different routes of administration	Change in end-tidal carbon dioxide percentage was increased for the three routes after $NaHCO_3$ bolus The peak level of end-tidal carbon dioxide was 2.5% for IO route, 3.0% for central IV and 1.9% for peripheral IV The IO route was slower to reach the peak effect. T_{max} was 30 s for IO route and was 21 s for central and peripheral IV ($p < .05$) The IO route had a longer duration of effect of 300 s compared with 180 s for central IV and with 240 s for peripheral IV ($p < .05$)	ARRIVE: 14/20
		Pollack Jr et al. [25]	Healthy animals	Group 1: IO normal saline 20 mL/kg ×2 Group 2: IO sodium bicarbonate 1 mEq/kg ×1 bolus undiluted Group 3: IO 10% sodium bicarbonate in D5W infusion at 80 mL/kg/d × 1 h Group 4: IO epinephrine 0.01 mg/kg ×1 h Group 5: IO epinephrine infusion 1 µg /kg/min ×1 h Group 6: IO dopamine infusion 10 µg /kg/min ×1 h *IO: left tibia *Control limb: right tibia	18 healthy male swine randomized to six groups ($n = 3$) to study long-term (3 months) effects on tibial bone marrow	- Mortality: 1 swine in group 1 died after 1 month from acute viral infection - Safety: no radiographic evidence of osteomyelitis and persistent bony defects. No histological changes. Marrow cellularity subjectively evaluated, one animal in group 1 showed a decrease when compared to control limb. No significant differences in the differential counts of white and red blood cell precursors between the control and the treated tibia	ARRIVE: 15.5/20
		Spivey et al. [27]	Cardiac arrest	Group 1: IO sodium bicarbonate 1 mEq/kg (n:6) Group 2: central IV sodium bicarbonate 1 mEq/kg ($n = 5$) Group 3: peripheral IV sodium bicarbonate	23 swine in induced ventricular fibrillation were resuscitated (CPR and ventilation) and randomized into four groups to compare different administration route	- Mortality: not mentioned - Relevant labs: pH obtained from femoral artery blood at different specific times following sodium bicarbonate administration. Group 1: 2 min pH: 7.66, 4 min pH: 7.68 Group 2: 2 min pH: 7.77, 4 min pH: 7.76	ARRIVE:14/20

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Antidote	Type of study	Source	Clinical context	Intervention	Subjects	Outcomes	Methodological quality
				1 mEq/kg ($n = 6$) Group 4: control ($n = 6$) *IO proximal tibia *Flushed with 20 mL of NS for groups 1, 2, and 3		Group 3: 2 min pH: 7.65, 4 min pH: 7.54 Group 4: 2 min pH: 7.42, 4 min pH: 7.36 (values from graph) At 2 min there was no significant pH difference a groups 1, 2, and 3 At 4 min groups 1 and 2 had significant higher pH values than group 3 and that difference remained until the end of the experiment No significant pH difference between groups 1 and 2 Same results when pH was obtained from left ventricle blood	
		Spivey et al. [28]	Healthy animals	Leg 1: IO sodium bicarbonate 1 mEq/kg ×1 rapid bolus (proximal tibia) ($n = 5$) Leg 2: IO NS 1 mL/kg ×1 rapid bolus in the opposite proximal tibia of the same swine ($n = 5$)	5 healthy domestic swine received IO sodium bicarbonate to assess potential bone changes	- Mortality: 30 d of daily observation $n =$ no mortality - Safety: no evidence of osteomyelitis or other complications. No triple-phase bone scans abnormalities. No gross and microscopic abnormalities Roentgenograph noted radiodensity of cortex at the needle puncture for $NaHCO_3$ leg for one animal	ARRIVE: 16/20
		Warren et al. [31]	Healthy animals	Group 1: sodium bicarbonate 1 mEq/kg injected into four sites: IO sites (3): proximal tibial or medial malleolar, distal femur and proximal humeral IV site (1): peripheral line ($n = 20$) *10 min stabilization after each injection for end-tidal CO_2 to return to baseline values	20 healthy Yorkshire–Landrace mix breed piglets in prospective study to assess sodium bicarbonate response to multiple IO sites compared with IV site	No difference between the sites for time to initial end-tidal CO_2 elevation (average: 12.8 s) and maximal end-tidal CO_2 elevation (average 1.09 kPa). In post hoc analysis: time of maximal end-tidal CO_2 increase was slower after IO (humerus: 25.0 s, femur: 26.4 seconds, malleolar 30.8 s, tibia 27.2 s) than IV injection (24.0 s) across all sites, but only statistically significantly lower for malleolar IO site versus IV and humerus sites	ARRIVE: 17/20
		Woodall et al. [32]	Healthy animals	Group 1: IO normal saline 20 mL/kg infusion ×20 min Group 2: IO sodium bicarbonate 1 mEq/kg ×1 bolus undiluted Group 3: IO 10% sodium bicarbonate in D5W infusion at 100 mL/kg/d × 1 h Group 4: IO epinephrine 0.01 mL/kg ×1 h Group 5: IO epinephrine infusion 1 µg /kg/min ×1 h Group 6: IO dopamine infusion 10 µg /kg/min ×1 h *IO: left tibia *Control limb: right tibia	18 healthy swine randomized into six groups ($n = 3$) to study long-term (6 months) potential linear bone growth effect	No difference of the linear growth between infused tibias and control tibias (right tibias) All tibias radiographed at 6 months revealed almost complete closure of epiphyses All defects (sclerotic border at IO insertion) had resolved at 6 months	ARRIVE: 16/20

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Antidote	Type of study	Source	Clinical context	Intervention	Subjects	Outcomes	Methodological quality
	Human cases	Carbajal et al. [35]	Pediatric dehydration	IO Sodium bicarbonate 1.4% 40 mL (1 mEq/kg =25 mL/kg) *IO left tibia plate	5-month-old infant with severe dehydration and hyperthermia (gastro-enteritis)	- Mortality: Infant survived - Relevant lab: pH increased at 7.31, Na: 162 (no pre sodium bicarbonate value) - Improvement of infant reactivity - Other IO medication: 35 mL colloids and 70 mL human albumin 20% - Safety: no inflammatory sign at IO access site at discharge (day 4)	Grade very low
		Ghirga et al. [40]	Pediatric septic shock	IO Sodium bicarbonate Unknown dose *IO left tibia	15-d-old (3.5 kg) infant	- Mortality: Infant died - Relevant labs: no information - Hemodynamics: increased heart rate - Other IO medication: Epinephrine 0.35 mL (1:10000)	Grade very low
		Herman et al. [42]	Poisoning + pediatric dehydration	IO Methylene blue 1% 1 mg/kg (0.3 mL) over 3–5 min IO Sodium bicarbonate 2 mEq/kg *IO proximal tibia	6-week-old infant with acquired methemoglobinemia	- Mortality: infant survived - Oximetry rose from 86% to 98–100% 8 min after antidote administration and cyanosis resolved - Relevant labs: methemoglobin concentration decreased from 29.3% to 8.2% (3 h after IO antidote administration) and 4.4% at 18.3 h (received a 2nd methylene blue dose via IV route at 7.8 h), acidosis at arrival (pH 7.21 before treatment) - Safety: no apparent side effect from the IO drugs administration	Grade very low
		Iserson [43]	Cardiac arrest	IO administration of sodium bicarbonate, Lidocaine, Atropine, and Vasopressors *IO medial malleolus *No information on doses	22 patients who arrived in ED in cardiac arrest in whom IV line was not established and had an IO needle inserted	- Mortality: 20 patients died in the ED, two others in the hospital within 24 h - Hemodynamics: in two patients, bradycardia responded to atropine within 1 min - Relevant labs: increase in pH in four of seven patients who received IO sodium bicarbonate. No possible evaluation for the three others - Safety: no evidence of complications from IO access	Grade very low
		McNamara et al. [53]	Cardiac arrest	IO Sodium bicarbonate 10 mL push *IO proximal tibia *No information on Sodium Bicarbonate concentration	3-month-old infant in cardiac arrest	- Mortality: patient died the next day - Epinephrine and Atropine given via endotracheal tube	Grade very low
		McNamara et al. [54]	Pediatric dehydration	IO Sodium bicarbonate 6 mEq/Kg *IO proximal tibia	7-month-old infant in severe dehydration	- Mortality: recovered uneventfully	Grade very low
		McNamara et al. [54]	Burn and cardiac arrest	IO Atropine and sodium bicarbonate *No information on IO localisation and on medication doses	Two infants (5 and 17 month-olds) in cardiac arrest found at a scene of house fire	- Mortality: both children died	Grade very low
		Mortensen et al. [50]	Poisoning	IO Sodium Bicarbonate 2.6 mEq/kg IO Dextrose 50% (unrecorded volume)	6-month-old infant intoxicated with 25 mg of encainide	- Mortality: Infant survived - Hemodynamics: impalpable femoral pulse to palpable pulse and BP 108/63 mmHg	GRADE VERY LOW

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Antidote	Type of study	Source	Clinical context	Intervention	Subjects	Outcomes	Methodological quality
				*IO proximal tibia		- Electrocardiographic variables: wide QRS (0.16 seconds) and ventricular fibrillation to sinus rhythm with normalization of QRS - Mental status: recovery of consciousness - Other relevant information: received IO phenytoin 20 mg/kg, fluids. CPR cardioversion and defibrillation were performed	
		Nafiu et al. [51]	Anaphylaxis Cardiac Arrest	IO Sodium Bicarbonate 8.4% 2 mL/kg diluted IO Dextrose 50% 1 mL/kg diluted *IO proximal tibia	1 infant in cardiac arrest following anaphylaxis	- Mortality: Infant survived - Hemodynamics: return to cardiac activity after first of 5 doses of IO epinephrine - Other IO medication: hydrocortisone 5 mg/kg, aminophylline 5 mg/kg and NS 50 mL	GRADE VERY LOW
		Ramet et al. [52]	Shock	IO Sodium Bicarbonate IO Atropine *No information about doses *IO tibial plate	800 g preterm infant	- Mortality: Infant died - Relevant labs: improvement of blood gases and electrolytes (no values) - Adverse effects: at autopsy no histological abnormalities at insertion site and bone marrow space - Other IO medication: Human albumin 10 mL/kg x 2 Dextrose 10% 130 mL/kg/day, epinephrine, vancomycin, cefotaxime, dobutamine and fentanyl	GRADE VERY LOW
		Salino et al. [54]	Pediatric dehydration and seizures	IO Sodium bicarbonate 1.4% (15 mL/kg) *IO antero-internal face of tibia	9-month-old infant with hyperthermia severe, seizure and dehydration (diarrhea since 48 hours)	- Mortality: no information - Hemodynamic improvement - Neurologic: seizures stopped - Adverse effects: no anomaly detected on tibial at day 15 of hospitalization - Also received by IO route: 10 mg/kg of phenobarbital, 20 mg/kg of acetylsalicylate, 30 mg/kg of cefotaxime	GRADE VERY LOW
			Pediatric dehydration	IO Sodium bicarbonate 1.4% (40 mL/kg) *IO superior extremity of right tibia	9-month-old infant with dehydration (gastroenteritis since 48 hours)	- Mortality: Infant survived - Relevant labs: improvement of the electrolytes results (Na: 156, K: 5.6, Cl: 126, then at 6 hours Na: 146, K: 3.8, Cl: 111) - Improvement of hemodynamics and mental status since no palpable pulse and comatose before medication. - Also received by IO route: 16 mL/kg of albumin 4%, 70 mg/kg of oxacillin and 7.5 mg/kg of amikacin and fluids (glucose 4.5% and NaCl 0.25%) - Adverse effects: no tibial growth problem detected 1 year later	

Appendix 2 – Ovid (Medline) search strategy

1. exp Acetylcysteine/
2. (acetylcysteine or "n-acetyl-l-cysteine" or "n acetyl l cysteine").tw.
3. (acebraus or acemuc or acetabs or acetyst or airbron or alveolex or azubronchin or bisolvon or bromuc or broncho?fips or broncholysin or broncoclar or codotussyl or cystamucil or (dampo adj mucopect) or durabronchal or eurespiran or exomuc or fabrol or fluimucil or fluprowit or frekatuss or genac or hoestil or (hustengetrank adj optipect) or jenacystein or jenapharm or lantamed or larylin or lindocetyl or (mercapturic adj acid) or muciteran or (muco adj sanigen) or mucomyst or mucosil or mucosol* or pectil or (sanigen adj muco) or siccoral or siran or solmucol or zambon).tw.
4. acetylcysteine.nm.
5. exp Atropine/
6. (atropin? or atropinol or cytospaz or anaspaz or hyoscyamine or lotomil or atropen).tw.
7. exp Calcium Chloride/
8. ((calcium adj2 chloride) or dianeal or extraneal or "elliot?s solution a" or "ringer?s injection").tw.
9. exp Calcium Gluconate/
10. ((calcium adj2 gluconate) or "gluconato calc" or ebucin or calciofon or calcivitol or calglucon or glucobiogen or glucal or calcipot or (rougier adj2 calcium)).tw.
11. exp Dantrolene/
12. (dantrolene or dantrium).tw.
13. exp Deferoxamine/
14. (deferoxamine or desferrioxamine or desferal).tw.
15. (D10W or D10%W or ((10% or 10?g) adj2 dextrose adj2 (water or solution))).tw.
16. (D50W or D50%W or ((50% or 50?g) adj2 dextrose adj2 (water or solution))).tw.
17. ((10% or 50%) adj2 dextrose).tw.
18. exp Diazepam/
19. Q3JTX2Q7TU.rn.
20. 7-chloro-1,3-dihydro-1-methyl-5-phenyl-2h-1,4-benzodiazepin-2-one.mp.
21. (diazemuls or diastat or apaurin or seduxen or faustan or azo-diazepam or novo-dipam or diazepam or sibazon or valium or vivol or stesolid or relanium).mp.
22. digoxin immune fab.mp.
23. (digifab or digibind).tw.
24. exp Diphenhydramine/
25. (dimedrol or benadryl or diphenhydramin? or diphenylhydramin? or benhydramin or benylin or benzhydramine or allerdryl or 2-diphenylmethoxy-N,N-dimethylethylamine or dormin or sinutab or balminil or unisom or allernix or calmex or dormex or dormiphen or ergodryl or insomnal or nadryl or nytol or sominex or diphenist).tw.
26. 8GTS82S83M.rn.
27. exp Ethanol/
28. (ethanol or alcohol or c2h5oh).tw.
29. exp Flumazenil/
30. 40P7XK9392.rn.
31. (flumazenil or flumazepil or "ro 151788" or anexate or "ro 15-1788" or romazicon or lanexat).tw.
32. exp Leucovorin/
33. (leu?ovorin or leu?ovorum or "folinic acid" or ((monosodium or sodium or calcium) adj folinate) or citrovorum or "5-formyltetrahydrofolate" or "5-formyltetrahydropteroylglutamate" or wellcovorin).tw.
34. Q573I9DVLP.rn.
35. fomepizole.mp.
36. 83LCM6L2BY.rn.
37. antizol.mp.
38. 4-methylpyrazole.mp.
39. 4-MP.mp.
40. exp Glucagon/
41. 9007-92-5.rn.
42. (glucagon or glucagen or glukagon or "hyperglycemic-glycogenolytic" or proglucagon or hg-factor).tw.
43. exp Hydroxocobalamin/
44. hydrox?cobalamin.tw.
45. (cyanokit or acti-B12 or B12 or B-12 or vitamin-B12 or "vitamin B12" or "vitamin B-12" or cobalamin?).tw.
46. Q40X8H422O.rn.
47. levocarnitine.mp.
48. carnitor.tw.
49. exp Methylene Blue/
50. T42P99266K.rn.
51. ("Phenothiazin-5-ium" or "3,7-bis(dimethylamino)-chloride").tw.
52. (((methylene or urolene or swiss) adj blue) or ((methylthionine or methylthioninium) adj chloride) or chromosmon).tw.
53. exp Naloxone/
54. (naloxon? or narcanti or "mrz 2593" or "mrz2593" or nalone or narcan or targin or suboxone).tw.
55. exp Phentolamine/
56. (phentolamine or z-max or regityn or r?gitine or fentolamin).tw.
57. Z468598HBV.rn.
58. exp Physostigmine/
59. (physostigmine or eserine).tw.
60. exp Pralidoxime Compounds/
61. 2-pyridine aldoxime methyl chloride.tw.
62. 2-PAM.tw.
63. protopam.tw.
64. pralidoxime.tw.
65. exp Protamines/
66. ("novomix 30" or humalog* or protamine*).tw.
67. exp Pyridoxine/
68. (pyridoxine or pralidoxime or diclectin or b-natal).tw.
69. exp Sodium Bicarbonate/
70. ((sodium adj bicarbonate) or ("carbonic acid" adj (sodium or monosodium)) or "baking soda" or "sodium hydrogen carbonate").tw.
71. (sodium adj (thiosulphate or thiosulfate)).tw.
72. exp Thiamine/
73. (thiamin? or "vitamin B1" or aneurin or vitathion or hemarexin or hormodausse or thiamiject).tw.
74. exp Vitamin K/
75. ("vitamin K?" or caltrate or phytomenadione or menaquinone or menadione).tw.
76. exp Prothrombin/
77. (prothrombin or (ii? adj coagulation adj factor) or (differentiation adj reversal adj factor) or "factor ii" or fibrinogenase or thrombase or thrombofort or thrombin-C or tropostasin or factor IIa or "E thrombin" or "beta-thrombin" or "gamma-thrombin" or beriplex or optaplex).tw.
78. exp Edetic Acid/ and (exp Calcium/ or calcium.mp.)
79. ((edta or tetracemate or "edetic acid" or ethylenediaminetetraacetic or edetate* or versenate* or versene* or chelat*) adj5 calcium).tw.
80. calcitetracemate.mp.
81. exp Insulin/
82. (insulin or iletin or novolin or hypurin or levemir or humulin or humalog).mp.
83. exp Fat Emulsions, Intravenous/
84. lipid rescue.mp.
85. (lipid adj3 emulsi*).mp.
86. (fat adj3 emulsi*).mp.
87. ((lipid or fat*) adj5 bolus).mp.
88. (lipid adj3 (resuscitat* or therap* or infus*)).mp.
89. (ILE adj5 (lipid* or emulsi* or fat*)).mp.
90. (IFE adj5 (lipid* or emulsi* or fat*)).mp.
91. (lipid adj3 sink*).mp.
92. (lipid adj3 sequest*).mp.
93. intravenous* lipid*.ti,ab,kw.
94. intralipid*.mp.
95. or/1-94
96. exp Infusions, Intraosseous/
97. ((intraosseous* or intra-osseous* or bone* or marrow) adj3 (inject* or infus* or deliver*)).mp.
98. 96 or 97
99. 95 and 98